UPSC ISS · STATISTICS PAPER-I · 2027 FORECAST

Probability

FORECAST · AI-GENERATED · NOT PREVIOUS-YEAR QUESTIONS

Every forecast problem in this unit, with its verified answer and a ~30-second exam shortcut.

Forecast problems here	**375** of 750 across the two units
Named forecast mocks	15, of exactly 25 questions each
Topics covered	11
Concepts covered	43
Layout of every entry	QUESTION → OPTIONS → ANSWER → EXAM SHORTCUT

These are not previous-year questions. Every stem, every option, every answer and every exam shortcut in this volume was written for this project as forecast practice for the 2027 attempt. Nothing here is reproduced from a UPSC paper and no official UPSC key exists for any of it.

Kept separate from the PYQ bank. The authentic 2018–2026 questions live in their own four volumes and in their own array inside the offline dashboard. Forecast material never enters a year, sectional, topic, subtopic or custom PYQ mock.

Answers were verified. Each key was worked out and independently re-checked during the build, but they remain a study aid, not an official key.

Contents

Named forecast mocks

MOCK	QUESTIONS	TOPICS
Probability Forecast Mock 12	25	11
Probability Forecast Mock 13	25	11
Probability Forecast Mock 14	25	11
Probability Forecast Mock 15	25	11

FORECAST · PROBABILITY

Characteristic Function, MGF and PGF

39 forecast problems · 4 concepts

1. FORECAST · FP-035 Characteristic Function

QUESTION

Which distribution has no moment generating function but has characteristic function $e^{-|t|}$?

OPTIONS

(a) Laplace

(b) Cauchy

(c) Lognormal

(d) Beta of the second kind

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) Cauchy

EXAM SHORTCUT (AI-generated explanation)

$e^{-|t|}$ is the standard Cauchy characteristic function, and the Cauchy has no MGF because it has no mean.

2. FORECAST · FP-069 Moment Generating Function

QUESTION

A random variable has $M(t) = (1/3 + (2/3)e^t)^5$. What is $\mathrm{Var}(X)$?

OPTIONS

(a) $10/3$

(b) $10/9$

(c) $5/9$

(d) $20/9$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $10/9$

EXAM SHORTCUT (AI-generated explanation)

Recognise $\mathrm{Binomial}(5, 2/3)$: variance $npq = 5(2/3)(1/3) = 10/9$.

3. **FORECAST · FP-070** Moment Generating Function

QUESTION

X is $N(1,4)$ and Y is $N(2,9)$, independent, and $Z = 2X - Y$. What is $\mathrm{P}(Z > 5)$?

OPTIONS

(a) 0.5

(b) 0.1587

(c) 0.0228

(d) 0.3085

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0.1587

EXAM SHORTCUT (AI-generated explanation)

Z is $N(0,25)$, so $\mathrm{P}(Z > 5) = \mathrm{P}(\text{standard normal} > 1) = 0.1587.$

4. **FORECAST · FP-071** Characteristic Function

QUESTION

Which characteristic function belongs to a distribution with no finite mean?

OPTIONS

(a) $\sin t/t$

(b) $1/(1+t^2)$

(c) $e^{-|t|}$

(d) $e^{-t^2/2}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) $e^{-|t|}$

EXAM SHORTCUT (AI-generated explanation)

$e^{-|t|}$ is the Cauchy characteristic function; it is not differentiable at 0, so no mean exists.

5. **FORECAST · FP-072** Compound Distributions

QUESTION

N is $\text{Poisson}(\lambda)$ and $X_1, X_2, \ldots$ are independent $\text{Bernoulli}(p)$ variables independent of N. The sum S of the first N of them follows

OPTIONS

(a) $\text{Binomial}(\lambda, p)$

(b) $\text{Poisson}(\lambda p)$

(c) $\text{Poisson}(\lambda)$

(d) a negative binomial distribution

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\text{Poisson}(\lambda p)$

EXAM SHORTCUT (AI-generated explanation)

> Compounding gives $\exp(\lambda(P_X(s) - 1)) = \exp(\lambda p(s - 1))$, the $\text{Poisson}(\lambda p)$ PGF.

6. **FORECAST · FP-073** Compound Distributions

QUESTION

N is geometric on $\{1, 2, \ldots\}$ with success probability p, and $X_1, X_2, \ldots$ are independent $\text{Exp}(\theta)$ variables independent of N. What is the MGF of the sum Z of the first N of them?

OPTIONS

(a) $p\theta/(p\theta - t)$

(b) $\theta/(\theta - t)$

(c) $(\theta/(\theta - t))^{1/p}$

(d) $p\theta/(\theta - t)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $p\theta/(p\theta - t)$

EXAM SHORTCUT (AI-generated explanation)

> Composing the geometric PGF with the exponential MGF gives $p\theta/(p\theta - t)$: Z is $\text{Exp}(p\theta)$.

7. **FORECAST · FP-074** Characteristic Function

QUESTION

Which statement about a characteristic function ϕ is FALSE?

OPTIONS

(a) $\phi(0) = 1$

(b) $|\phi(t)|$ <= 1 for all t

(c) ϕ is unchanged by a change of origin

(d) $\phi(-t)$ is the complex conjugate of $\phi(t)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) ϕ is unchanged by a change of origin

EXAM SHORTCUT (AI-generated explanation)

> Shifting X to X + b multiplies ϕ by e^{itb}, so the characteristic function is not origin-invariant.

8. **FORECAST · FP-075** Moment Generating Function

QUESTION

A random variable has $M(t) = e^{t^2}$. What is $\mathrm{Var}(X)$?

OPTIONS

(a) 1

(b) 2

(c) $1/2$

(d) 4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 2

EXAM SHORTCUT (AI-generated explanation)

> The normal MGF is $\exp(\mu t + \sigma^2 t^2/2)$, so $\sigma^2/2 = 1$ and the variance is 2.

9. **FORECAST · FP-076** Moment Generating Function

QUESTION

A random variable has $M(t) = (1 - t/2)^{-3}$ for $t < 2$. What are its mean and variance?

OPTIONS

(a) $3/2$ and $3/4$

(b) 6 and 12

(c) $3/2$ and $3/2$

(d) 3 and $3/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $3/2$ and $3/4$

EXAM SHORTCUT (AI-generated explanation)

> Γ with shape 3 and rate 2: mean $3/2$, variance $3/4$.

10. **FORECAST · FP-169** Compound Distributions

QUESTION

The number of customers N is $\mathrm{Poisson}(\lambda)$ and each spends an amount with mean μ and variance σ^2, independently. What is the variance of the total spend?

OPTIONS

(a) $\lambda\sigma^2$

(b) $\lambda(\sigma^2 + \mu^2)$

(c) $\lambda\mu^2$

(d) $\lambda^2\sigma^2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\lambda(\sigma^2 + \mu^2)$

EXAM SHORTCUT (AI-generated explanation)

> For a compound Poisson, $\mathrm{Var}(S) = \lambda\,\mathrm{E}(X^2) = \lambda(\sigma^2 + \mu^2)$.

11. FORECAST · FP-176 Moment Generating Function

QUESTION

A random variable has $M(t) = 1/(1 - t^2)$ for $|t| < 1$. What is $\mathrm{Var}(X)$?

OPTIONS

(a) 1

(b) 2

(c) $1/2$

(d) 4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 2

EXAM SHORTCUT (AI-generated explanation)

The series $1 + t^2 + t^4 + \ldots$ has coefficient 1 at t^2, so $\mathrm{E}(X^2) = 2! = 2$ and the mean is 0.

12. FORECAST · FP-177 Probability Generating Function

QUESTION

A probability generating function is $\mathrm{P}(s) = s/(2 - s)$. What are the mean and the variance?

OPTIONS

(a) 2 and 2

(b) 1 and 1

(c) 2 and 4

(d) $1/2$ and $1/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 2 and 2

EXAM SHORTCUT (AI-generated explanation)

Rewrite as $(s/2)/(1 - s/2)$: geometric on {1,2,...} with $p = 1/2$, mean 2 and variance $q/p^2 = 2$.

13. FORECAST · FP-178 Characteristic Function

QUESTION

A characteristic function is $\phi(t) = \cos^2 t$. What is the distribution of X?

OPTIONS

(a) plus or minus 1 with probability $1/2$ each

(b) 0 with probability $1/2$ and plus or minus 2 with probability $1/4$ each

(c) $U(-1, 1)$

(d) plus or minus 2 with probability $1/2$ each

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0 with probability $1/2$ and plus or minus 2 with probability $1/4$ each

EXAM SHORTCUT (AI-generated explanation)

$\cos^2 t = (1 + \cos 2t)/2$, which is the characteristic function of a mass $1/2$ at 0 and $1/4$ at each of 2 and -2.

14. FORECAST · FP-179 Moment Generating Function

QUESTION

A random variable has $M(t) = \exp(2(e^t - 1))$ times $(1/2 + e^t/2)^3$. What is $\mathrm{Var}(X)$?

OPTIONS

(a) 2

(b) $11/4$

(c) $7/2$

(d) 3

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $11/4$

EXAM SHORTCUT (AI-generated explanation)

The product is $\mathrm{Poisson}(2)$ plus an independent $\mathrm{Binomial}(3, 1/2)$: variance $2 + 3/4 = 11/4$.

15. **FORECAST · FP-180** Characteristic Function

QUESTION

$X_1, \ldots, X_n$ are independent standard Cauchy variables. What is the characteristic function of their sample mean?

OPTIONS

(a) $e^{-|t|/n}$

(b) $e^{-|t|}$

(c) $e^{-n|t|}$

(d) $e^{-|t|/\sqrt{n}}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $e^{-|t|}$

EXAM SHORTCUT (AI-generated explanation)

Each term contributes $e^{-|t|/n}$ and there are n of them, so the product is $e^{-|t|}$ again.

16. **FORECAST · FP-181** Moment Generating Function

QUESTION

A random variable has $M(t) = (e^t - 1)/t$. What is $\mathrm{E}(X^2)$?

OPTIONS

(a) $1/2$

(b) $1/3$

(c) $1/6$

(d) $1/12$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/3$

EXAM SHORTCUT (AI-generated explanation)

This is the $U(0, 1)$ MGF, so $\mathrm{E}(X^2) = 1/3$.

17. **FORECAST · FP-182** Compound Distributions

QUESTION

N is $\mathrm{Poisson}(\lambda)$, the X_i are independent $\mathrm{Exp}(\theta)$ variables, and S is the sum of the first N of them. What is $\mathrm{Var}(S)$?

OPTIONS

(a) λ/θ^2

(b) $2\lambda/\theta^2$

(c) λ/θ

(d) λ^2/θ^2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $2\lambda/\theta^2$

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{Var}(S) = \lambda\,\mathrm{E}(X^2) = \lambda(2/\theta^2)$.

18. **FORECAST · FP-183** Moment Generating Function

QUESTION

The cumulant generating function is $K(t) = \lambda(e^t - 1)$. Every cumulant equals

OPTIONS

(a) λ^r

(b) λ

(c) $r!\lambda$

(d) zero beyond the second

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) λ

EXAM SHORTCUT (AI-generated explanation)

Every derivative of λe^t at $t = 0$ is λ, so all cumulants equal λ.

19. **FORECAST · FP-184** Characteristic Function

QUESTION

Which statement is FALSE?

OPTIONS

(a) equal moment generating functions on an interval around 0 imply equal distributions

(b) equal moment sequences imply equal distributions

(c) the characteristic function always exists

(d) the characteristic function is uniformly continuous

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) equal moment sequences imply equal distributions

EXAM SHORTCUT (AI-generated explanation)

> The lognormal has infinitely many distinct competitors with the same moments, so moments alone do not determine a law.

20. **FORECAST · FP-185** Characteristic Function

QUESTION

The characteristic function of X is real-valued for every t if and only if

OPTIONS

(a) X is non-negative

(b) X is symmetric about 0

(c) X has a finite mean

(d) X is bounded

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) X is symmetric about 0

EXAM SHORTCUT (AI-generated explanation)

> ϕ is real exactly when $\phi(t)$ equals its conjugate $\phi(-t)$, i.e. when X and -X have the same law.

21. FORECAST · FP-186 Probability Generating Function

QUESTION

For a probability generating function $\mathrm{P}(s), \mathrm{P}(X \text{ is even})$ equals

OPTIONS

(a) $\mathrm{P}(1/2)$

(b) $[\mathrm{P}(1)+\mathrm{P}(-1)]/2$

(c) $\mathrm{P}(0)$

(d) $[\mathrm{P}(1)-\mathrm{P}(-1)]/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $[\mathrm{P}(1)+\mathrm{P}(-1)]/2$

EXAM SHORTCUT (AI-generated explanation)

> Averaging P at $s=1$ and $s=-1$ cancels the odd terms and doubles the even ones.

22. FORECAST · FP-244 Compound Distributions

QUESTION

N is $\mathrm{Binomial}(n,p)$ and $X_1, X_2, \ldots$ are independent $\mathrm{Bernoulli}(r)$ variables independent of N. The sum of the first N of them follows

OPTIONS

(a) $\mathrm{Binomial}(n, pr)$

(b) $\mathrm{Binomial}(np, r)$

(c) $\mathrm{Poisson(npr)}$

(d) $\mathrm{Binomial}(n, p+r-pr)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\mathrm{Binomial}(n, pr)$

EXAM SHORTCUT (AI-generated explanation)

> Composing the two PGFs gives $(q+p(1-r+rs))^n = (1-pr+prs)^n$.

23. **FORECAST · FP-245** Compound Distributions

QUESTION

N is geometric on {0,1,2,...} with success probability p and the X_i are independent $\mathrm{Exp}(\theta)$. The randomly stopped sum S is

OPTIONS

(a) $\mathrm{Exp}(p\theta)$

(b) 0 with probability p, and otherwise $\mathrm{Exp}(p\theta)$

(c) $\Gamma(1/p, \theta)$

(d) $\mathrm{Exp}(\theta/p)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0 with probability p, and otherwise $\mathrm{Exp}(p\theta)$

EXAM SHORTCUT (AI-generated explanation)

$M_S(t) = p(\theta - t)/(p\theta - t) = p + q$ times $p\theta/(p\theta - t)$: an atom at 0 plus an exponential.

24. **FORECAST · FP-246** Compound Distributions

QUESTION

N is $\mathrm{Poisson}(\lambda)$ and the X_i are independent $\mathrm{Poisson}(\mu)$. What are $\mathrm{E}(S)$ and $\mathrm{Var}(S)$ for the randomly stopped sum?

OPTIONS

(a) $\lambda\mu$ and $\lambda\mu$

(b) $\lambda\mu$ and $\lambda\mu(1+\mu)$

(c) $\lambda\mu$ and $\lambda\mu^2$

(d) $\lambda + \mu$ and $\lambda + \mu$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\lambda\mu$ and $\lambda\mu(1+\mu)$

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{E}(S) = \lambda\mu$ and $\mathrm{Var}(S) = \lambda\,\mathrm{E}(X^2) = \lambda(\mu + \mu^2)$.

25. FORECAST · FP-247 Compound Distributions

QUESTION

N is $\mathrm{Poisson}(\lambda)$ and each X_i is 1 or -1 with probability $1/2$. What is the characteristic function of the randomly stopped sum?

OPTIONS

(a) $\exp(\lambda(\cos t - 1))$

(b) $\exp(\lambda(e^{it} - 1))$

(c) $\cos^{\lambda} t$

(d) $\exp(-\lambda t^2/2)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\exp(\lambda(\cos t - 1))$

EXAM SHORTCUT (AI-generated explanation)

> The summand has characteristic function $\cos t$, and compounding gives $\exp(\lambda(\cos t - 1))$.

26. FORECAST · FP-248 Compound Distributions

QUESTION

For a randomly stopped sum S of independent identically distributed X_i with N independent of them, $\mathrm{Var}(S)$ equals

OPTIONS

(a) $\mathrm{E}(N)\,\mathrm{Var}(X)$

(b) $\mathrm{E}(N)\,\mathrm{Var}(X) + \mathrm{Var}(N)[\mathrm{E}(X)]^2$

(c) $\mathrm{Var}(N)\,\mathrm{Var}(X)$

(d) $\mathrm{E}(N)\,\mathrm{E}(X^2)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\mathrm{E}(N)\,\mathrm{Var}(X) + \mathrm{Var}(N)[\mathrm{E}(X)]^2$

EXAM SHORTCUT (AI-generated explanation)

> The variance decomposition gives a within-term part and a between-count part.

27. FORECAST · FP-249 Compound Distributions

QUESTION

N is $\mathrm{Poisson}(\lambda)$ and each of the N items is labelled type A independently with probability p. The counts of type A and type B are

OPTIONS

(a) negatively correlated

(b) independent $\mathrm{Poisson}(\lambda p)$ and $\mathrm{Poisson}(\lambda(1-p))$

(c) $\mathrm{Binomial}(N, p)$ and $\mathrm{Binomial}(N, 1-p)$

(d) positively correlated

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) independent $\mathrm{Poisson}(\lambda p)$ and $\mathrm{Poisson}(\lambda(1-p))$

EXAM SHORTCUT (AI-generated explanation)

Poisson thinning splits a Poisson stream into two INDEPENDENT Poisson streams.

28. FORECAST · FP-250 Compound Distributions

QUESTION

The X_i are independent $\mathrm{Exp}(1)$ and N is $\mathrm{Poisson}(\lambda)$, independent of them. What is the variance of the randomly stopped sum?

OPTIONS

(a) λ

(b) 2λ

(c) λ^2

(d) $\lambda + \lambda^2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 2λ

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{Var}(S) = \lambda\,\mathrm{E}(X^2) = 2\lambda.$

29. FORECAST · FP-253 Characteristic Function

QUESTION

Statement I: if two variables have the same characteristic function they have the same distribution. Statement II: the inversion theorem shows that a characteristic function determines its distribution uniquely. Choose the correct code.

OPTIONS

(a) both true, II explains I

(b) both true, II does not explain I

(c) I true, II false

(d) I false, II true

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) both true, II explains I

EXAM SHORTCUT (AI-generated explanation)

> Both statements are true and the inversion theorem is precisely the reason for the first.

30. FORECAST · FP-254 Characteristic Function

QUESTION

Statement I: if the characteristic function of X + Y factorises as the product of those of X and Y, then X and Y are independent. Statement II: taking X standard Cauchy and $Y = X$ gives a factorisation without independence. Choose the correct code.

OPTIONS

(a) both true, II explains I

(b) both true, II does not explain I

(c) I true, II false

(d) I false, II true

ANSWER (AI-generated forecast item — no official UPSC key exists)

(d) I false, II true

EXAM SHORTCUT (AI-generated explanation)

> The factorisation is necessary but not sufficient; the Cauchy pair X and $Y = X$ is the standard counterexample.

31. FORECAST · FP-301 Characteristic Function

QUESTION

X and Y are independent $\text{Poisson}(\lambda)$. What is the characteristic function of X - Y?

OPTIONS

(a) $\exp(2\lambda(\cos t - 1))$

(b) $\exp(\lambda(e^{it} - 1))$

(c) $\exp(-2\lambda t^2)$

(d) $\exp(2\lambda(e^{it} - 1))$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\exp(2\lambda(\cos t - 1))$

EXAM SHORTCUT (AI-generated explanation)

Multiplying the characteristic functions of X and -Y gives $\exp(\lambda(e^{it} + e^{-it} - 2)) = \exp(2\lambda(\cos t - 1))$.

32. FORECAST · FP-317 Moment Generating Function

QUESTION

A random variable has $M(t) = (1 - t)^{-1/2}$ for $t < 1$. What are its mean and variance?

OPTIONS

(a) $1/2$ and $1/2$

(b) $1/2$ and $1/4$

(c) 1 and 2

(d) 2 and 4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/2$ and $1/2$

EXAM SHORTCUT (AI-generated explanation)

Γ with shape $1/2$ and rate 1: mean $1/2$ and variance $1/2$.

33. **FORECAST · FP-318** Moment Generating Function

QUESTION

A random variable has $M(t) = 1/3 + (2/3)e^{2t}$. What values does it take, and with what probabilities?

OPTIONS

(a) 0 and 2 with probabilities $1/3$ and $2/3$

(b) 1 and 2 with probabilities $1/3$ and $2/3$

(c) 0 and 1 with probabilities $1/3$ and $2/3$

(d) the value 2 with probability one

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0 and 2 with probabilities $1/3$ and $2/3$

EXAM SHORTCUT (AI-generated explanation)

> Each term ce^{at} in an MGF is an atom of size c at the point a.

34. **FORECAST · FP-319** Characteristic Function

QUESTION

Which of the following is NOT a characteristic function?

OPTIONS

(a) $\cos t$

(b) $e^{-|t|}$

(c) e^{-t^2}

(d) e^{-t^4}

ANSWER (AI-generated forecast item — no official UPSC key exists)

(d) e^{-t^4}

EXAM SHORTCUT (AI-generated explanation)

> By Marcinkiewicz's theorem, $\exp$(a polynomial) is a characteristic function only if the polynomial has degree at most 2.

35. FORECAST · FP-320 Characteristic Function

QUESTION

A characteristic function is $\phi(t) = \sin t/t$. What is $\mathrm{Var}(X)$?

OPTIONS

(a) $1/3$

(b) $1/12$

(c) 1

(d) $2/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/3$

EXAM SHORTCUT (AI-generated explanation)

> $\sin t/t$ is the $U(-1, 1)$ characteristic function, whose variance is $4/12 = 1/3$.

36. FORECAST · FP-321 Characteristic Function

QUESTION

For a characteristic function ϕ of X, the squared modulus of $\phi(t)$ is the characteristic function of

OPTIONS

(a) 2X

(b) X plus an independent copy of X

(c) X minus an independent copy of X

(d) X^2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) X minus an independent copy of X

EXAM SHORTCUT (AI-generated explanation)

> The squared modulus equals $\phi(t)\phi(-t)$, the characteristic function of the difference of two independent copies.

37. **FORECAST · FP-322** Moment Generating Function

QUESTION

X is $\text{Binomial}(n, p)$. What is the moment generating function of n - X?

OPTIONS

(a) $(q + pe^{-t})^n$

(b) $(p + qe^t)^n$

(c) $e^{nt}(q + pe^t)^n$

(d) $(q + pe^t)^{-n}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $(p + qe^t)^n$

EXAM SHORTCUT (AI-generated explanation)

n - X is $\text{Binomial}(n, q)$, so its MGF is $(p + qe^t)^n$.

38. **FORECAST · FP-323** Probability Generating Function

QUESTION

For a probability generating function $\text{P}(s)$, the k-th derivative at $s = 1$ equals

OPTIONS

(a) $\text{E}(X^k)$

(b) $\text{E}[X(X-1)\dots(X-k+1)]$

(c) k! times $\text{P}(X = k)$

(d) the variance when $k = 2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\text{E}[X(X-1)\dots(X-k+1)]$

EXAM SHORTCUT (AI-generated explanation)

Differentiating k times and setting $s = 1$ produces the k-th factorial moment.

39. **FORECAST · FP-370** Moment Generating Function

QUESTION

Which statement about the moment generating function is TRUE?

OPTIONS

(a) it exists for every random variable

(b) when it is finite on an open interval around 0 it determines the distribution

(c) it is always bounded by 1

(d) it is real only for symmetric variables

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) when it is finite on an open interval around 0 it determines the distribution

EXAM SHORTCUT (AI-generated explanation)

Finiteness on a neighbourhood of the origin makes the MGF analytic there and determines the law uniquely.

FORECAST · PROBABILITY

Classical and Axiomatic Probability

25 forecast problems · 4 concepts

1. FORECAST · FP-001 Axioms and Elementary Probability

QUESTION

A, B and C are independent events with $\mathrm{P}(A)=1/2, \mathrm{P}(B)=1/3$ and $\mathrm{P}(C)=1/4$. What is $\mathrm{P}(\text{at least one of them occurs})$?

OPTIONS

(a) $1/4$

(b) $3/4$

(c) $11/24$

(d) $13/24$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $3/4$

EXAM SHORTCUT (AI-generated explanation)

Complement: $1-(1/2)(2/3)(3/4)=1-1/4=3/4.$

2. FORECAST · FP-002 Combinatorial Probability

QUESTION

Ten persons sit at random around a round table. What is the probability that exactly 3 persons sit between A and B?

OPTIONS

(a) $1/9$

(b) $2/9$

(c) $3/10$

(d) $1/5$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $2/9$

EXAM SHORTCUT (AI-generated explanation)

Fix A. B is equally likely to take any of the 9 remaining seats; 2 of them leave exactly 3 people between.

3. FORECAST · FP-003 Geometric Probability

QUESTION

Two points are chosen independently and uniformly on $(0, 1)$. What is the probability that the distance between them is less than $1/2$?

OPTIONS

(a) $1/2$

(b) $1/4$

(c) $3/4$

(d) $2/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) $3/4$

EXAM SHORTCUT (AI-generated explanation)

> The complement $|X - Y| \geq 1/2$ is two corner triangles of leg $1/2$: area $2(1/2)(1/2)^2 = 1/4$. Answer $3/4$.

4. FORECAST · FP-004 Geometric Probability

QUESTION

X and Y are independent $U(0, 1)$ variables. What is the probability that the equation $t^2 + 2$Xt + $Y = 0$ has real roots?

OPTIONS

(a) $1/2$

(b) $1/3$

(c) $2/3$

(d) $1/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/3$

EXAM SHORTCUT (AI-generated explanation)

> Real roots need the discriminant $4X^2 - 4Y \geq 0$, i.e. $Y \leq X^2$. Area under $y = x^2$ on $(0, 1)$ is $1/3$.

5. FORECAST · FP-005 Combinatorial Probability

QUESTION

Five letters are placed at random into five addressed envelopes, one per envelope. What is the probability that no letter goes into its own envelope?

OPTIONS

(a) $11/30$

(b) $1/5$

(c) $19/30$

(d) $1/120$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $11/30$

EXAM SHORTCUT (AI-generated explanation)

> The number of derangements of 5 objects is 44, so the probability is $44/120 = 11/30$.

6. FORECAST · FP-006 Bounds on Probabilities

QUESTION

If $\mathrm{P}(A) = 0.7$ and $\mathrm{P}(B) = 0.6$, the smallest possible value of $\mathrm{P}(A \text{ intersect B})$ is

OPTIONS

(a) 0

(b) 0.3

(c) 0.42

(d) 0.6

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0.3

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{P}(A \text{ union B}) \leq 1$ forces $\mathrm{P}(A \text{ and B}) \geq 0.7 + 0.6 - 1 = 0.3$.

7. **FORECAST · FP-101** Combinatorial Probability

QUESTION

A fair die is rolled four times. What is the probability that the four outcomes are strictly increasing?

OPTIONS

(a) $15/1296$

(b) $1/54$

(c) $5/324$

(d) $1/24$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $15/1296$

EXAM SHORTCUT (AI-generated explanation)

> Each 4-subset of the six faces gives exactly one increasing order: $\binom{6}{4}/6^4 = 15/1296$.

8. **FORECAST · FP-102** Combinatorial Probability

QUESTION

Eight people including one couple sit at random in a row. What is the probability that the couple sit together and neither of them is at an end?

OPTIONS

(a) $1/4$

(b) $5/28$

(c) $3/14$

(d) $5/14$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $5/28$

EXAM SHORTCUT (AI-generated explanation)

> Five interior adjacent slots, two internal orders, 6! for the rest: $5(2)(6!)/8! = 10/56 = 5/28$.

9. FORECAST · FP-103 Geometric Probability

QUESTION

A point is chosen uniformly in the unit disc. What is the probability that it is nearer to the centre than to the boundary?

OPTIONS

(a) $1/2$

(b) $1/4$

(c) $1/\pi$

(d) $1/\sqrt{2}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/4$

EXAM SHORTCUT (AI-generated explanation)

Nearer to the centre means $r < 1 - r$, i.e. $r < 1/2$, an area ratio of $(1/2)^2 = 1/4$.

10. FORECAST · FP-104 Combinatorial Probability

QUESTION

An integer is drawn uniformly from 1 to 100. What is the probability that it is divisible by 2, 3 or 5?

OPTIONS

(a) 0.70

(b) 0.74

(c) 0.76

(d) 0.66

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0.74

EXAM SHORTCUT (AI-generated explanation)

Inclusion-exclusion: $50 + 33 + 20 - 16 - 10 - 6 + 3 = 74$, so the probability is 0.74.

11. **FORECAST · FP-105** Combinatorial Probability

QUESTION

A hand of 5 cards is dealt from a standard pack of 52. What is the probability that exactly two suits appear?

OPTIONS

(a) $\binom{4}{2}\binom{26}{5}/\binom{52}{5}$

(b) $\binom{4}{2}[\binom{26}{5} - 2\binom{13}{5}]/\binom{52}{5}$

(c) $2\binom{26}{5}/\binom{52}{5}$

(d) $\binom{4}{2}\binom{13}{5}^2/\binom{52}{5}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\binom{4}{2}[\binom{26}{5} - 2\binom{13}{5}]/\binom{52}{5}$

EXAM SHORTCUT (AI-generated explanation)

> Choose the two suits, count all hands inside them, then subtract the two single-suit hands.

12. **FORECAST · FP-262** Combinatorial Probability

QUESTION

Three balls are placed at random into three cells. What is the probability that exactly one cell is left empty?

OPTIONS

(a) $1/3$

(b) $2/3$

(c) $2/9$

(d) $4/9$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $2/3$

EXAM SHORTCUT (AI-generated explanation)

> Choose the empty cell in 3 ways and split 3 balls into the other 2 with neither empty: $3(2^3 - 2)/27 = 18/27$.

13. FORECAST · FP-263 Combinatorial Probability

QUESTION

What is the probability that at least two of four people were born in the same calendar month?

OPTIONS

(a) $55/96$

(b) $41/96$

(c) $1/3$

(d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $41/96$

EXAM SHORTCUT (AI-generated explanation)

Complement: $1 - 12(11)(10)(9)/12^4 = 1 - 11880/20736 = 41/96.$

14. FORECAST · FP-264 Combinatorial Probability

QUESTION

In a random permutation of $\{1, \ldots, n\}$, what is the probability that 1 and 2 lie in the same cycle?

OPTIONS

(a) $1/n$

(b) $1/2$

(c) $2/n$

(d) $1/(n-1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/2$

EXAM SHORTCUT (AI-generated explanation)

By symmetry the answer is exactly $1/2$ for every $n \geq 2$.

15. FORECAST · FP-265 Axioms and Elementary Probability

QUESTION

Two dice are rolled repeatedly. What is the probability that a sum of 7 appears before a sum of 8?

OPTIONS

(a) $1/2$

(b) $6/11$

(c) $5/11$

(d) $6/36$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $6/11$

EXAM SHORTCUT (AI-generated explanation)

Condition on the first decisive roll: $6/(6+5) = 6/11$.

16. FORECAST · FP-266 Geometric Probability

QUESTION

A stick is broken at two independent uniform random points. What is the probability that the three pieces form a triangle?

OPTIONS

(a) $1/2$

(b) $1/4$

(c) $1/3$

(d) $1/8$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/4$

EXAM SHORTCUT (AI-generated explanation)

The triangle inequality fails unless both break points lie on opposite sides of the midpoint and are within $1/2$: probability $1/4$.

17. FORECAST · FP-268 Axioms and Elementary Probability

QUESTION

Two fair coins are tossed. A is the event that the first is a head, B that the second is a head and C that both show the same face. Which statement is correct?

OPTIONS

(a) A, B and C are mutually independent

(b) they are pairwise independent only, and $\mathrm{P}(C \text{ given A and B}) = 1$

(c) they are not pairwise independent

(d) $\mathrm{P}(A \text{ and B and C}) = 1/8$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) they are pairwise independent only, and $\mathrm{P}(C \text{ given A and B}) = 1$

EXAM SHORTCUT (AI-generated explanation)

Each pair is independent with probability $1/4$, but $\mathrm{P}(A \text{ and B and C}) = 1/4$, not $1/8$.

18. FORECAST · FP-354 Axioms and Elementary Probability

QUESTION

Can a σ field over a finite sample space contain exactly 6 sets?

OPTIONS

(a) yes, for a suitable sample space

(b) no, because a finite σ field always has 2^k members

(c) yes, but only if the space has 6 points

(d) only if the space is infinite

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) no, because a finite σ field always has 2^k members

EXAM SHORTCUT (AI-generated explanation)

A finite σ field is generated by a partition into k atoms, so it has exactly 2^k members; 6 is not a power of 2.

19. **FORECAST · FP-355** Axioms and Elementary Probability

QUESTION

Two events A and B are mutually exclusive with $\mathrm{P}(A) > 0$ and $\mathrm{P}(B) > 0$. Then they are

OPTIONS

(a) independent

(b) never independent

(c) independent if and only if $\mathrm{P}(A) + \mathrm{P}(B) = 1$

(d) independent if and only if $\mathrm{P}(A) = \mathrm{P}(B)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) never independent

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{P}(A \text{ and } \mathrm{B}) = 0$ while $\mathrm{P}(A)\,\mathrm{P}(B) > 0$, so the product rule fails for every such pair.

20. **FORECAST · FP-356** Axioms and Elementary Probability

QUESTION

Two fair coins are tossed. A is the event that the first is a head, B that the second is a head and C that exactly one head appears. In Bernstein's example these three events are

OPTIONS

(a) mutually independent

(b) pairwise independent but not mutually independent

(c) not pairwise independent

(d) disjoint

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) pairwise independent but not mutually independent

EXAM SHORTCUT (AI-generated explanation)

Each pair has probability $1/4$, but $\mathrm{P}(A \text{ and } \mathrm{B} \text{ and } \mathrm{C}) = 0$ while the product of the three is $1/8$.

21. FORECAST · FP-360 Geometric Probability

QUESTION

Bertrand's chord paradox shows that "the probability that a random chord of a circle is longer than the inscribed equilateral triangle's side" can be

OPTIONS

(a) only $1/3$

(b) $1/2, 1/3$ or $1/4$ depending on how the chord is randomised

(c) only $1/2$

(d) undefined in every formulation

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/2, 1/3$ or $1/4$ depending on how the chord is randomised

EXAM SHORTCUT (AI-generated explanation)

> The random-endpoint, random-radius and random-midpoint constructions give $1/3, 1/2$ and $1/4$ respectively.

22. FORECAST · FP-361 Bounds on Probabilities

QUESTION

Boole's inequality states that the probability of a union is at most the sum of the probabilities. The corresponding Bonferroni lower bound for the intersection of n events is

OPTIONS

(a) the sum of the probabilities minus $(n-1)$

(b) the product of the probabilities

(c) the sum of the probabilities

(d) 1 minus the product of the probabilities

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) the sum of the probabilities minus $(n-1)$

EXAM SHORTCUT (AI-generated explanation)

> Complementing Boole bounds the intersection below by the sum of the $\mathrm{P}(A_i)$ minus $(n-1)$.

23. FORECAST · FP-362 Axioms and Elementary Probability

QUESTION

Which implication is correct?

OPTIONS

(a) $\mathrm{P}(A) = 0$ implies that A is the empty set

(b) $\mathrm{P}(A) = 1$ implies that A is the whole sample space

(c) $\mathrm{P}(A) = 0$ does not imply that A is empty

(d) $\mathrm{P}(A) = \mathrm{P}(B)$ implies $A = B$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) $\mathrm{P}(A) = 0$ does not imply that A is empty

EXAM SHORTCUT (AI-generated explanation)

A continuous variable hits any fixed point with probability 0, yet that single-point event is not empty.

24. FORECAST · FP-363 Axioms and Elementary Probability

QUESTION

For a sequence of events increasing to A, the continuity property of a probability measure gives

OPTIONS

(a) $\mathrm{P}(A_n)$ increases to $\mathrm{P}(A)$

(b) $\mathrm{P}(A_n)$ decreases to $\mathrm{P}(A)$

(c) nothing without an extra finiteness condition

(d) $\mathrm{P}(A_n)$ equals $\mathrm{P}(A)$ for all large n

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\mathrm{P}(A_n)$ increases to $\mathrm{P}(A)$

EXAM SHORTCUT (AI-generated explanation)

Countable additivity gives continuity from below directly for any probability measure.

25. **FORECAST · FP-375** Axioms and Elementary Probability

QUESTION

In the ISS objective paper each question carries 2.5 marks with a penalty of one third of that for a wrong answer. The expected gain from a pure random guess among four options is

OPTIONS

(a) positive

(b) zero

(c) negative

(d) $2.5/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) zero

EXAM SHORTCUT (AI-generated explanation)

$0.25(2.5) - 0.75(0.8333) = 0.625 - 0.625 = 0.$

FORECAST · PROBABILITY

Conditional Probability and Bayes Theorem

20 forecast problems · 2 concepts

1. FORECAST · FP-007 Bayes Theorem

QUESTION

Machines M_1, M_2 and M_3 produce 50%, 30% and 20% of the output with defect rates 2%, 3% and 4%. An item drawn at random is found defective. What is the probability that it came from M_3?

OPTIONS

(a) $8/27$

(b) $4/27$

(c) $10/27$

(d) $9/27$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $8/27$

EXAM SHORTCUT (AI-generated explanation)

Weights $0.5(0.02) : 0.3(0.03) : 0.2(0.04) = 10$: 9 : 8, so the answer is $8/27$.

2. FORECAST · FP-008 Total Probability

QUESTION

Urn 1 has 2 white and 3 black balls, urn 2 has 3 white and 2 black, urn 3 has 4 white and 1 black. An urn is chosen at random and one ball is drawn. What is the probability that it is white?

OPTIONS

(a) $8/15$

(b) $2/5$

(c) $3/5$

(d) $4/5$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) $3/5$

EXAM SHORTCUT (AI-generated explanation)

Equal urn totals, so just average the white counts: $(2 + 3 + 4)/(3 \times 5) = 9/15 = 3/5$.

3. FORECAST · FP-009 Bayes Theorem

QUESTION

A disease has prevalence 1%. A test has sensitivity 95% and specificity 90%. Given a positive test, what is the probability that the person has the disease?

OPTIONS

(a) 0.95

(b) 0.0876

(c) 0.105

(d) 0.19

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0.0876

EXAM SHORTCUT (AI-generated explanation)

> True positives $0.01(0.95) = 0.0095$ against false positives $0.99(0.10) = 0.099$; ratio $0.0095/0.1085 = 19/217$.

4. FORECAST · FP-010 Bayes Theorem

QUESTION

Of four coins, three are fair and one is two-headed. One coin is chosen at random and tossed three times, giving three heads. What is the probability that the two-headed coin was chosen?

OPTIONS

(a) $8/11$

(b) $1/4$

(c) $3/11$

(d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $8/11$

EXAM SHORTCUT (AI-generated explanation)

> Weights $(1/4)(1)$ against $(3/4)(1/8) = 3/32$; ratio $(8/32)/(11/32) = 8/11$.

5. FORECAST · FP-106 Total Probability

QUESTION

Two fair dice are thrown. What is P(sum = 7 given that at least one die shows 3)?

OPTIONS

(a) $1/6$

(b) $2/11$

(c) $1/11$

(d) $1/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $2/11$

EXAM SHORTCUT (AI-generated explanation)

The conditioning event has 11 outcomes, of which $(3, 4)$ and $(4, 3)$ give a sum of 7.

6. FORECAST · FP-108 Bayes Theorem

QUESTION

Three dice are available: a fair one, one with faces 1,2,3,4,6,6 and one with faces 1,2,3,6,6,6. One is chosen at random and rolled twice, giving two sixes. What is the probability that the fair die was chosen?

OPTIONS

(a) $1/3$

(b) $1/14$

(c) $1/9$

(d) $2/7$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/14$

EXAM SHORTCUT (AI-generated explanation)

Likelihoods $1/36, 4/36$ and $9/36$ give posterior weights 1 : 4 : 9, so the fair die has probability $1/14$.

7. FORECAST · FP-109 Bayes Theorem

QUESTION

30% of mail is spam. A word W appears in 60% of spam and 5% of legitimate mail. Two independent messages each contain W. What is the probability that both are spam?

OPTIONS

(a) $(36/43)^2$

(b) $36/43$

(c) 0.36

(d) $(18/43)^2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $(36/43)^2$

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{P}(\text{spam} \mid W) = 0.18/(0.18 + 0.035) = 36/43$, and independence squares it.

8. FORECAST · FP-110 Bayes Theorem

QUESTION

An urn holds 3 balls; the number of white balls is 0, 1, 2 or 3 with equal prior probability. Two draws with replacement are both white. What is the probability that all three balls are white?

OPTIONS

(a) $9/14$

(b) $3/4$

(c) $1/2$

(d) $9/16$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $9/14$

EXAM SHORTCUT (AI-generated explanation)

Likelihood $(k/3)^2$ gives weights 0, 1, 4, 9, so the answer is $9/14$.

9. FORECAST · FP-111 Bayes Theorem

QUESTION

A bit 0 or 1 is sent with equal probability over a channel that flips each transmitted bit independently with probability 0.1. The same bit is sent twice and the receiver observes 1 then 0. What is the probability that the bit sent was 1?

OPTIONS

(a) 0.9

(b) 0.1

(c) $1/2$

(d) $0.81/0.82$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) $1/2$

EXAM SHORTCUT (AI-generated explanation)

Both hypotheses give the same likelihood $0.9(0.1)$, so the data are uninformative and the posterior equals the prior.

10. FORECAST · FP-112 Bayes Theorem

QUESTION

The success probability p of a coin is $U(0,1)$. The coin is tossed twice and both tosses are heads. What is the probability that a third toss is a head?

OPTIONS

(a) $2/3$

(b) $3/4$

(c) $1/2$

(d) $4/5$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $3/4$

EXAM SHORTCUT (AI-generated explanation)

Laplace's rule: $\mathrm{E}(p^3)/\mathrm{E}(p^2) = (1/4)/(1/3) = 3/4$, which is $(k+1)/(n+2)$ with $k = n = 2$.

11. FORECAST · FP-267 Bayes Theorem

QUESTION

Three boxes contain two gold coins, two silver coins and one of each. A box is chosen at random and one coin drawn is gold. What is the probability that the other coin in that box is gold?

OPTIONS

(a) $1/2$

(b) $2/3$

(c) $1/3$

(d) $3/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $2/3$

EXAM SHORTCUT (AI-generated explanation)

> Count coins, not boxes: two of the three gold coins live in the two-gold box.

12. FORECAST · FP-269 Bayes Theorem

QUESTION

There are four doors and one prize. You pick one door; the host then opens a door with no prize among the other three. You switch to one of the two remaining unopened doors. What is the probability that you win?

OPTIONS

(a) $1/4$

(b) $3/8$

(c) $1/2$

(d) $1/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $3/8$

EXAM SHORTCUT (AI-generated explanation)

> Your first door holds the prize with probability $1/4$; otherwise the prize is behind one of the two you might switch to, so $3/4$ times $1/2$.

13. FORECAST · FP-270 Bayes Theorem

QUESTION

Prevalence is 0.1 and two conditionally independent tests each have sensitivity 0.9 and specificity 0.8. Both are positive. What is the posterior probability of disease?

OPTIONS

(a) $9/13$

(b) 0.81

(c) $1/2$

(d) $9/22$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $9/13$

EXAM SHORTCUT (AI-generated explanation)

Weights $0.1(0.81) = 0.081$ against $0.9(0.04) = 0.036$, giving $81/117 = 9/13$.

14. FORECAST · FP-271 Bayes Theorem

QUESTION

Given θ, X is exponential with rate θ, and the prior on θ is $\mathrm{Exp}(1)$. What is the posterior distribution of θ given $X = x$?

OPTIONS

(a) $\mathrm{Exp}(1 + x)$

(b) Γ with shape 2 and rate 1 + x

(c) Γ with shape 1 and rate x

(d) $\mathrm{Beta}(1, x)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) Γ with shape 2 and rate 1 + x

EXAM SHORTCUT (AI-generated explanation)

The posterior is proportional to $\theta e^{-\theta(1+x)}$, which is $\Gamma(2, 1 + x)$.

15. FORECAST · FP-272 Bayes Theorem

QUESTION

"$\mathrm{P}(\text{evidence given innocent}) = 0.001$, therefore $\mathrm{P}(\text{innocent given evidence}) = 0.001$." This reasoning

OPTIONS

(a) is valid by Bayes' theorem

(b) is valid if the prior is uniform

(c) ignores the prior and the probability of the evidence under guilt

(d) is valid asymptotically

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) ignores the prior and the probability of the evidence under guilt

EXAM SHORTCUT (AI-generated explanation)

> Bayes needs the prior odds and the likelihood under the alternative; swapping the conditioning is the prosecutor's fallacy.

16. FORECAST · FP-273 Bayes Theorem

QUESTION

Urn I has 1 red and 1 blue ball; urn II has 2 red balls. An urn is chosen at random, a red ball is drawn and replaced, and a red ball is drawn again. What is the probability that urn II was chosen?

OPTIONS

(a) $2/3$

(b) $4/5$

(c) $3/4$

(d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $4/5$

EXAM SHORTCUT (AI-generated explanation)

> Likelihoods $(1/2)^2 = 1/4$ against 1, so the posterior is $1/(1+1/4) = 4/5$.

17. FORECAST · FP-357 Total Probability

QUESTION

Which statement about conditional probability is TRUE?

OPTIONS

(a) $\mathrm{P}(A \text{ given B})$ plus $\mathrm{P}(A \text{ given the complement of B})$ equals 1

(b) $\mathrm{P}(A \text{ given B})$ plus $\mathrm{P}(\text{the complement of A given B})$ equals 1

(c) $\mathrm{P}(A \text{ given B})$ is never smaller than $\mathrm{P}(A)$

(d) $\mathrm{P}(A \text{ given B})$ equals $\mathrm{P}(B \text{ given A})$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\mathrm{P}(A \text{ given B})$ plus $\mathrm{P}(\text{the complement of A given B})$ equals 1

EXAM SHORTCUT (AI-generated explanation)

For a fixed conditioning event the conditional probability is a genuine probability measure, so complements within it sum to 1.

18. FORECAST · FP-358 Bayes Theorem

QUESTION

In the Monty Hall problem the host opens a door AT RANDOM among the two you did not choose, and it happens to reveal a goat. Switching now wins with probability

OPTIONS

(a) $2/3$

(b) $1/2$

(c) $1/3$

(d) $3/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/2$

EXAM SHORTCUT (AI-generated explanation)

An uninformed host who happens to miss the car provides symmetric evidence, so the two remaining doors are equally likely.

19. FORECAST · FP-359 Bayes Theorem

QUESTION

A family has two children with independent and equally likely sexes. Given that at least one is a boy, what is the probability that both are boys?

OPTIONS

(a) $1/2$

(b) $1/3$

(c) $1/4$

(d) $2/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/3$

EXAM SHORTCUT (AI-generated explanation)

> The conditioning set has three equally likely outcomes and only one has two boys.

20. FORECAST · FP-364 Total Probability

QUESTION

Cards are drawn one at a time without replacement from a well-shuffled pack. The unconditional probability that the third card is an ace is

OPTIONS

(a) $1/13$

(b) $4/49$

(c) $1/17$

(d) smaller than the probability for the first card

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/13$

EXAM SHORTCUT (AI-generated explanation)

> By symmetry every position is equally likely to hold an ace, so the answer is the initial proportion $4/52$.

FORECAST · PROBABILITY

Distribution Functions and Transformations

31 forecast problems · 5 concepts

1. FORECAST · FP-033 Functions of Random Variables

QUESTION

X is $U(-1,1)$ and $Y = X^2$. What is the density of Y on $(0,1)$?

OPTIONS

(a) $2\sqrt{y}$

(b) $1/(2\sqrt{y})$

(c) 1

(d) $1/\sqrt{y}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/(2\sqrt{y})$

EXAM SHORTCUT (AI-generated explanation)

$F_Y(y) = \mathrm{P}(|X| < \sqrt{y}) = \sqrt{y}$, so the density is $1/(2\sqrt{y})$.

2. FORECAST · FP-058 Probability Integral Transform

QUESTION

X is $U(0,1)$ and $Y = -\log X$. Then Y follows

OPTIONS

(a) $\mathrm{Exp}(1)$

(b) an exponential with mean 2

(c) $U(0,\infty)$

(d) $\Gamma(2,1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\mathrm{Exp}(1)$

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{P}(Y > y) = \mathrm{P}(X < e^{-y}) = e^{-y}$, the standard exponential survival function.

3. **FORECAST · FP-094** Distribution Function Properties

QUESTION

$F(x)$ is 0 for $x < 0, \alpha x$ for $0 \leq x < 1$ and 1 for $x \geq 1$. This is a valid distribution function if and only if

OPTIONS

(a) $\alpha = 1$

(b) $0 \leq \alpha \leq 1$

(c) $\alpha > 0$

(d) $\alpha \geq 1$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $0 \leq \alpha \leq 1$

EXAM SHORTCUT (AI-generated explanation)

> F only has to be non-decreasing and bounded by 1, so $\alpha \geq 0$ and $\alpha \leq 1$; a jump at $x = 1$ is perfectly legal.

4. **FORECAST · FP-095** Distribution Function Properties

QUESTION

$F(x)$ is 0 for $x < 0, 1/4 + \alpha x$ for $0 \leq x < 1$ and 1 for $x \geq 1$. What is the admissible range of α?

OPTIONS

(a) $0 \leq \alpha \leq 3/4$

(b) $\alpha = 3/4$

(c) $0 \leq \alpha \leq 1$

(d) $-1/4 \leq \alpha \leq 3/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $0 \leq \alpha \leq 3/4$

EXAM SHORTCUT (AI-generated explanation)

> Non-decreasing needs $\alpha \geq 0$ and $F(1-) = 1/4 + \alpha \leq 1$ needs $\alpha \leq 3/4$.

5. **FORECAST · FP-096** Probability Integral Transform

QUESTION

F is a continuous distribution function and X has that distribution. Then $-2\log F(X)$ follows

OPTIONS

(a) $\text{Exp}(1)$

(b) an exponential with mean 2

(c) $U(0, 2)$

(d) $N(0, 1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) an exponential with mean 2

EXAM SHORTCUT (AI-generated explanation)

$F(X)$ is $U(0, 1)$, and $-2\log U$ is exponential with mean 2.

6. **FORECAST · FP-097** Distribution Function Properties

QUESTION

F_1 and F_2 are distribution functions. Which of the following need NOT be a distribution function?

OPTIONS

(a) the product of F_1 and F_2

(b) $F_1 + F_2 - F_1F_2$

(c) the average of F_1 and F_2

(d) $F_1 - F_2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(d) $F_1 - F_2$

EXAM SHORTCUT (AI-generated explanation)

A difference of distribution functions can be negative and does not tend to 1, so it fails.

7. FORECAST · FP-098 Distribution Function Properties

QUESTION

$F(x) = 1/2 + (1/\pi)\arctan x$. What is $\mathrm{P}(X > 1)$?

OPTIONS

(a) $1/2$

(b) $1/4$

(c) $3/4$

(d) $1/\pi$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/4$

EXAM SHORTCUT (AI-generated explanation)

This is the standard Cauchy distribution function; $F(1) = 1/2 + 1/4 = 3/4$, so the upper tail is $1/4$.

8. FORECAST · FP-099 Distribution Function Properties

QUESTION

How many jump points does $F(x) = 1 - e^{-x}$ for $x \geq 0$, and 0 otherwise, have?

OPTIONS

(a) none

(b) one, at $x = 0$

(c) infinitely many

(d) one, at $x = 1$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) none

EXAM SHORTCUT (AI-generated explanation)

$F(0) = 0 = F(0-)$, so there is no jump anywhere; the exponential distribution function is continuous.

9. FORECAST · FP-100 Distribution Function Properties

QUESTION

Statement I: a distribution function must be continuous. Statement II: a distribution function must be right-continuous. Which is correct?

OPTIONS

(a) Both statements are true

(b) I is false, II is true

(c) I is true, II is false

(d) Both statements are false

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) I is false, II is true

EXAM SHORTCUT (AI-generated explanation)

Discrete variables have jumps, so continuity fails; right-continuity is part of the definition.

10. FORECAST · FP-209 Mixed Distributions

QUESTION

$F(x)$ is 0 for $x < 0, (x+1)/4$ for $0 \le x < 1$ and 1 for $x \ge 1$. What are $\mathrm{P}(X=0)$ and $\mathrm{P}(X=1)$?

OPTIONS

(a) 0 and 0

(b) $1/4$ and $1/2$

(c) $1/4$ and $1/4$

(d) 0 and $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/4$ and $1/2$

EXAM SHORTCUT (AI-generated explanation)

The jumps are $F(0) - F(0-) = 1/4$ and $F(1) - F(1-) = 1 - 1/2 = 1/2$.

11. FORECAST · FP-210 Mixed Distributions

QUESTION

For $F(x) = (x+1)/4$ on $0 \leq x < 1$, with $F = 0$ below 0 and 1 from 1 onwards, what is $\mathrm{E}(X)$?

OPTIONS

(a) $1/2$

(b) $5/8$

(c) $3/4$

(d) $9/16$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $5/8$

EXAM SHORTCUT (AI-generated explanation)

The atom at 1 contributes $1/2$, and the continuous density $1/4$ on $(0, 1)$ contributes $1/8$.

12. FORECAST · FP-211 Distribution Function Properties

QUESTION

$F(x)$ is 0 for $x < 0, \alpha + (1 - \alpha)x$ for $0 \leq x < 1$ and 1 for $x \geq 1$. This is a valid distribution function if and only if

OPTIONS

(a) $\alpha = 0$

(b) $0 \leq \alpha \leq 1$

(c) $\alpha > 0$

(d) $0 < \alpha < 1$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $0 \leq \alpha \leq 1$

EXAM SHORTCUT (AI-generated explanation)

α is the atom at 0, so any value in 0 to 1 works; the endpoints give a degenerate atom and a pure uniform.

13. FORECAST · FP-212 Distribution Function Properties

QUESTION

For the logistic distribution function $F(x) = 1/(1 + e^{-x})$, the density equals

OPTIONS

(a) $F(1 - F)$

(b) F^2

(c) e^{-x}

(d) 1 - F

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $F(1 - F)$

EXAM SHORTCUT (AI-generated explanation)

Differentiating gives $e^{-x}/(1 + e^{-x})^2 = F(1 - F)$.

14. FORECAST · FP-213 Probability Integral Transform

QUESTION

X is $\text{Bernoulli}(1/2)$ with distribution function F. What values does $F(X)$ take?

OPTIONS

(a) it is $U(0, 1)$

(b) $1/2$ and 1 with probability $1/2$ each

(c) 0 and 1

(d) 0 and $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/2$ and 1 with probability $1/2$ each

EXAM SHORTCUT (AI-generated explanation)

$F(0) = 1/2$ and $F(1) = 1$, so $F(X)$ is a two-point variable, not uniform.

15. **FORECAST · FP-214** Probability Integral Transform

QUESTION

U is $U(0,1)$. Which transform gives the standard Cauchy distribution?

OPTIONS

(a) $-\log(1-U)$

(b) $\tan(\pi(U-1/2))$

(c) the ceiling of $\log U$ divided by $\log q$

(d) the square root of U

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\tan(\pi(U-1/2))$

EXAM SHORTCUT (AI-generated explanation)

> The Cauchy quantile function is $\tan(\pi(u-1/2))$.

16. **FORECAST · FP-215** Hazard and Survival

QUESTION

The survival function is $S(x)=(1+x)^{-2}$ for $x>0$. What is the hazard rate?

OPTIONS

(a) $2/(1+x)$

(b) $2(1+x)^{-3}$

(c) $1/(1+x)$

(d) the constant 2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $2/(1+x)$

EXAM SHORTCUT (AI-generated explanation)

> $h=f/S=2(1+x)^{-3}/(1+x)^{-2}=2/(1+x)$.

17. FORECAST · FP-216 Functions of Random Variables

QUESTION

X has distribution function F. Then $G(x) = F(\sqrt{x})$ for $x \geq 0$ and 0 otherwise is the distribution function of X^2

OPTIONS

(a) always

(b) only if X is non-negative with probability one

(c) only if X is symmetric

(d) never

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) only if X is non-negative with probability one

EXAM SHORTCUT (AI-generated explanation)

In general $\mathrm{P}(X^2 \leq x) = F(\sqrt{x}) - F(-\sqrt{x}$ minus), so the left tail must be empty for the formula to hold.

18. FORECAST · FP-217 Distribution Function Properties

QUESTION

g is strictly increasing and X is continuous. Which statement is correct?

OPTIONS

(a) both the mean and the median of $g(X)$ are g of the corresponding quantity

(b) only the median transforms in that way

(c) only the mean transforms in that way

(d) neither transforms in that way

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) only the median transforms in that way

EXAM SHORTCUT (AI-generated explanation)

Quantiles commute with strictly increasing maps; expectations do not, by Jensen's inequality.

19. FORECAST · FP-218 Distribution Function Properties

QUESTION

The set of discontinuity points of a distribution function is

OPTIONS

(a) finite

(b) at most countable

(c) possibly uncountable

(d) empty

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) at most countable

EXAM SHORTCUT (AI-generated explanation)

Jumps larger than $1/n$ number at most n, and a countable union of finite sets is countable.

20. FORECAST · FP-235 Functions of Random Variables

QUESTION

X is $U(0, 1)$ and $Y = 4X(1 - X)$. What is the density of Y on $(0, 1)$?

OPTIONS

(a) $1/(4\sqrt{1-y})$

(b) $1/(2\sqrt{1-y})$

(c) $1/(2\sqrt{y})$

(d) $2(1-y)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/(2\sqrt{1-y})$

EXAM SHORTCUT (AI-generated explanation)

Two branches x = (1 plus or minus $\sqrt{1-y}$)/2, each contributing $1/(4\sqrt{1-y})$.

21. FORECAST · FP-236 Functions of Random Variables

QUESTION

X is $U(0,1)$ and $Y = 4X(1-X)$. Then Y follows

OPTIONS

(a) $\mathrm{Beta}(1/2, 1)$

(b) $\mathrm{Beta}(1, 1/2)$

(c) $U(0,1)$

(d) the arcsine distribution

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\mathrm{Beta}(1, 1/2)$

EXAM SHORTCUT (AI-generated explanation)

> The density is proportional to $y^0(1-y)^{-1/2}$, which is $\mathrm{Beta}(1, 1/2)$.

22. FORECAST · FP-237 Functions of Random Variables

QUESTION

X is $U(0,1)$ and $Y = 4X(1-X)$. What is $\mathrm{P}(Y > 3/4)$?

OPTIONS

(a) $1/4$

(b) $1/2$

(c) $3/4$

(d) $1/\sqrt{2}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/2$

EXAM SHORTCUT (AI-generated explanation)

> $4 \times (1-x) > 3/4$ gives $1/4 < x < 3/4$, an interval of length $1/2$.

23. FORECAST · FP-345 Distribution Function Properties

QUESTION

$F(x) = 1 - x^{-2}$ for $x \geq 1$. What are the mean and the variance?

OPTIONS

(a) 2 and infinite

(b) 2 and 4

(c) 1 and infinite

(d) both infinite

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 2 and infinite

EXAM SHORTCUT (AI-generated explanation)

The density is $2x^{-3}$, so the mean is 2 but the second moment integral diverges logarithmically.

24. FORECAST · FP-346 Distribution Function Properties

QUESTION

$F(x) = \exp(-e^{-x})$ for all real x. What is the median?

OPTIONS

(a) 0

(b) $-\log(\log 2)$

(c) $\log 2$

(d) 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $-\log(\log 2)$

EXAM SHORTCUT (AI-generated explanation)

Solving $\exp(-e^{-m}) = 1/2$ gives $e^{-m} = \log 2$, so $m = -\log(\log 2)$.

25. FORECAST · FP-347 Functions of Random Variables

QUESTION

Y is $\mathrm{Exp}(1)$ and $X = \sqrt{Y}$. What are the distribution function and the mean of X?

OPTIONS

(a) $1 - e^{-x^2}$ and $\sqrt{\pi}/2$

(b) $1 - e^{-x}$ and 1

(c) $1 - e^{-x^2}$ and 1

(d) $1 - e^{-\sqrt{x}}$ and 2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1 - e^{-x^2}$ and $\sqrt{\pi}/2$

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{P}(X \le x) = \mathrm{P}(Y \le x^2) = 1 - e^{-x^2}$; the mean is $\mathrm{E}(\mathrm{sqrt\ Y}) = \Gamma(3/2) = \sqrt{\pi}/2.$

26. FORECAST · FP-348 Probability Integral Transform

QUESTION

The quantile function is $Q(u) = u^{1/\alpha}$ for $\alpha > 0$. The distribution is

OPTIONS

(a) $\mathrm{Beta}(\alpha, 1)$

(b) $\mathrm{Beta}(1, \alpha)$

(c) $\mathrm{Exp}(\alpha)$

(d) $U(0, 1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\mathrm{Beta}(\alpha, 1)$

EXAM SHORTCUT (AI-generated explanation)

Inverting gives $F(x) = x^\alpha$ on $(0, 1)$, which is the $\mathrm{Beta}(\alpha, 1)$ distribution function.

27. FORECAST · FP-349 Distribution Function Properties

QUESTION

$F(x) = (1/2)(1 + x/\sqrt{1 + x^2})$. Which statement is correct?

OPTIONS

(a) it is not a distribution function

(b) it is a distribution function with mean 0 and infinite variance

(c) it is a distribution function with mean 0 and variance 1

(d) it is a distribution function whose mean is undefined

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) it is a distribution function with mean 0 and infinite variance

EXAM SHORTCUT (AI-generated explanation)

The density is $(1/2)(1 + x^2)^{-3/2}$, whose tail behaves like x^{-3}: the mean converges but the variance does not.

28. FORECAST · FP-350 Mixed Distributions

QUESTION

$F(x)$ is 0 for $x < 0, 1/2$ for $0 \leq x < 1$, and $1 - (1/2)e^{-(x-1)}$ for $x \geq 1$. What are the atoms and $\mathrm{E}(X)$?

OPTIONS

(a) an atom of $1/2$ at 0 only, and $\mathrm{E}(X) = 1$

(b) atoms at 0 and 1, and $\mathrm{E}(X) = 3/2$

(c) no atoms, and $\mathrm{E}(X) = 1$

(d) an atom at 0, and $\mathrm{E}(X) = 2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) an atom of $1/2$ at 0 only, and $\mathrm{E}(X) = 1$

EXAM SHORTCUT (AI-generated explanation)

$F(1) = 1/2 = F(1-)$, so the only atom is $1/2$ at the origin; the continuous half is $1 + \mathrm{Exp}(1)$ with mean 2.

29. **FORECAST · FP-351** Functions of Random Variables

QUESTION

X is continuous and symmetric about 0 with distribution function F. For $x \geq 0$, the distribution function of $|X|$ is

OPTIONS

(a) $F(x)$

(b) $2F(x) - 1$

(c) $1 - F(x)$

(d) $F(x)^2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $2F(x) - 1$

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{P}(|X| \leq x) = F(x) - F(-x) = F(x) - (1 - F(x)) = 2F(x) - 1.$

30. **FORECAST · FP-352** Distribution Function Properties

QUESTION

F and G are distribution functions. The composition x maps to $F(G(x))$ is a distribution function

OPTIONS

(a) always

(b) if and only if $F(0) = 0$ and $F(1) = 1$

(c) never

(d) if and only if G is continuous

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) if and only if $F(0) = 0$ and $F(1) = 1$

EXAM SHORTCUT (AI-generated explanation)

> As x tends to minus ∞, G tends to 0 and the composition tends to $F(0)$; the limits must be 0 and 1.

31. FORECAST · FP-353 Probability Integral Transform

QUESTION

For a general distribution function F, define the generalised inverse as the infimum of the x with $F(x)$ at least u. If U is $U(0,1)$, the variable obtained by applying it to U has distribution function

OPTIONS

(a) F

(b) the uniform distribution function

(c) F, only if F is continuous

(d) F, only if F is strictly increasing

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) F

EXAM SHORTCUT (AI-generated explanation)

The generalised inverse transform works for EVERY distribution function, discrete or mixed included.

FORECAST · PROBABILITY

Laws of Large Numbers and Central Limit Theorem

25 forecast problems · 3 concepts

1. FORECAST · FP-088 Central Limit Theorem

QUESTION

For 300 independent $U(0,2)$ variables, what is the approximate value of $\mathrm{P}(\mathrm{sum} > 310)$?

OPTIONS

(a) 0.1587

(b) 0.0228

(c) 0.3085

(d) 0.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.1587

EXAM SHORTCUT (AI-generated explanation)

> Mean 300 and variance $300(1/3) = 100$, so $z = 10/10 = 1$ and the tail is 0.1587.

2. FORECAST · FP-089 Central Limit Theorem

QUESTION

1200 round-off errors are independent $U(-0.5, 0.5)$. What is the approximate value of $\mathrm{P}(|\sum| \leq 20)$?

OPTIONS

(a) 0.6826

(b) 0.9544

(c) 0.9973

(d) 0.50

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0.9544

EXAM SHORTCUT (AI-generated explanation)

> Variance $1200/12 = 100$, so 20 is two standard deviations and the central probability is 0.9544.

3. FORECAST · FP-090 Weak and Strong Laws

QUESTION

Statement I: the strong law of large numbers for independent and identically distributed variables requires a finite variance. Statement II: the sample mean of independent $\mathrm{Cauchy}(0,1)$ variables converges to 0 in probability. Which is correct?

OPTIONS

(a) Both statements are true

(b) Both statements are false

(c) I is true, II is false

(d) I is false, II is true

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) Both statements are false

EXAM SHORTCUT (AI-generated explanation)

Kolmogorov's strong law needs only a finite mean, and the Cauchy sample mean is again Cauchy, so it converges to nothing.

4. FORECAST · FP-091 Central Limit Theorem

QUESTION

For a random sample from $U(-3,3)$, the large-sample distribution of $\sqrt{n}$ times the sample mean is

OPTIONS

(a) $N(0,1)$

(b) $N(0,3)$

(c) $N(0,9)$

(d) $U(-3,3)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $N(0,3)$

EXAM SHORTCUT (AI-generated explanation)

The variance is $36/12=3$, so $\sqrt{n}\bar{X}$ tends to $N(0,3)$.

5. FORECAST · FP-092 Central Limit Theorem

QUESTION

$X_1, X_2, \ldots$ are independent and identically distributed with mean 0 and variance σ^2. Which normalisation converges in distribution to a non-degenerate limit?

OPTIONS

(a) the sum divided by n

(b) the sum divided by $\sqrt{n}$

(c) the sum divided by $n^{1/4}$

(d) the sum divided by n^2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the sum divided by $\sqrt{n}$

EXAM SHORTCUT (AI-generated explanation)

Only the $\sqrt{n}$ scaling keeps the variance constant; the answer is the central limit theorem normalisation.

6. FORECAST · FP-093 Central Limit Theorem

QUESTION

X is $\mathrm{Poisson}(100)$. Using the central limit theorem without a continuity correction, $\mathrm{P}(X \leq 110)$ is approximately

OPTIONS

(a) $\Phi(1)$

(b) $\Phi(0.1)$

(c) $\Phi(10)$

(d) $\Phi(1.1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\Phi(1)$

EXAM SHORTCUT (AI-generated explanation)

Mean and variance are both 100, so the standard deviation is 10 and $z = 10/10 = 1$.

7. FORECAST · FP-200 Central Limit Theorem

QUESTION

$X_1, \ldots, X_{100}$ are independent $\mathrm{Exp}(1)$. What is the approximate value of $\mathrm{P}(\mathrm{sum} > 110)$?

OPTIONS

(a) 0.1587

(b) 0.0228

(c) 0.3085

(d) 0.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.1587

EXAM SHORTCUT (AI-generated explanation)

Mean 100 and variance 100, so $z = 1$ and the tail is 0.1587.

8. FORECAST · FP-201 Central Limit Theorem

QUESTION

X_i are independent with density 2x on $(0, 1)$. Then $\sqrt{n}$ times (the sample mean minus $2/3$) tends to $N(0, v)$. What is v?

OPTIONS

(a) $1/12$

(b) $1/18$

(c) $1/9$

(d) $2/9$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/18$

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{E}(X^2) = 1/2$ and the mean is $2/3$, so the variance is $1/2 - 4/9 = 1/18$.

9. FORECAST · FP-202 Delta Method

QUESTION

For a $\text{Bernoulli}(p)$ sample, $\sqrt{n}$ times ($\bar{X}(1-\bar{X})$ minus $p(1-p)$) tends to $N(0, v)$. At $p = 1/2$, what is v?

OPTIONS

(a) $1/4$

(b) $1/16$

(c) 0

(d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 0

EXAM SHORTCUT (AI-generated explanation)

> The delta-method factor is $g'(p)^2 pq = (1-2p)^2 pq$, which vanishes at $p = 1/2$.

10. FORECAST · FP-203 Weak and Strong Laws

QUESTION

X_k are independent with common mean μ and $\text{Var}(X_k) = k$. Does the sample mean converge to μ in probability?

OPTIONS

(a) yes, by Chebyshev

(b) yes, by the strong law

(c) not guaranteed; the variance of the sample mean tends to $1/2$

(d) yes, for normal variables

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) not guaranteed; the variance of the sample mean tends to $1/2$

EXAM SHORTCUT (AI-generated explanation)

> $\text{Var}(\bar{X}) = (1/n^2)$ times the sum of $k = (n+1)/(2n)$, which tends to $1/2$ rather than 0.

11. **FORECAST · FP-204** Central Limit Theorem

QUESTION

A fair coin is tossed 400 times. What is the approximate probability that the number of heads lies between 180 and 220 inclusive?

OPTIONS

(a) 0.6827

(b) 0.9544

(c) 0.9973

(d) 0.75

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0.9544

EXAM SHORTCUT (AI-generated explanation)

> Mean 200 and standard deviation 10, so the interval is the mean plus or minus 2σ: 0.9544.

12. **FORECAST · FP-205** Weak and Strong Laws

QUESTION

For a sample from $U(0,\theta)$, which estimator is strongly consistent for θ by the strong law alone?

OPTIONS

(a) the sample mean

(b) twice the sample mean

(c) half the sample mean

(d) the square of the sample mean

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) twice the sample mean

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{E}(X) = \theta/2$, so the sample mean converges almost surely to $\theta/2$ and twice it converges to θ.

13. FORECAST · FP-206 Delta Method

QUESTION

X_i are independent with mean μ and variance σ^2. Then $\sqrt{n}$ times (the square of the sample mean minus μ^2) tends to $N(0, v)$. What is v?

OPTIONS

(a) σ^2

(b) $4\mu^2\sigma^2$

(c) $2\mu\sigma^2$

(d) σ^4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $4\mu^2\sigma^2$

EXAM SHORTCUT (AI-generated explanation)

Δ method with $g(x) = x^2$: $v = g'(\mu)^2\sigma^2 = 4\mu^2\sigma^2$.

14. FORECAST · FP-207 Central Limit Theorem

QUESTION

A central limit theorem for independent but not identically distributed summands holds under

OPTIONS

(a) finite variances alone

(b) the Lindeberg or Liapounov conditions

(c) no circumstances

(d) identical means

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the Lindeberg or Liapounov conditions

EXAM SHORTCUT (AI-generated explanation)

Finite variances are not enough; the Lindeberg condition rules out any single term dominating the sum.

15. **FORECAST · FP-208** Central Limit Theorem

QUESTION

X_i are independent $\mathrm{Poisson}(1)$ and $n = 100$. Using the central limit theorem without a continuity correction, the probability that the sum is at most 90 is approximately

OPTIONS

(a) $\Phi(-1)$

(b) $\Phi(-0.1)$

(c) $\Phi(-10)$

(d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\Phi(-1)$

EXAM SHORTCUT (AI-generated explanation)

Mean 100 and variance 100, so $z = -10/10 = -1$ and the answer is $\Phi(-1)$.

16. **FORECAST · FP-258** Weak and Strong Laws

QUESTION

Statement I: if the X_n are independent and identically distributed with $\mathrm{E}|X_1|$ infinite, then the sample mean cannot converge almost surely to a finite constant. Statement II: an infinite first absolute moment forces the limit superior of $|X_n|/n$ to be infinite almost surely. Choose the correct code.

OPTIONS

(a) both true, II explains I

(b) both true, II does not explain I

(c) I true, II false

(d) I false, II true

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) both true, II explains I

EXAM SHORTCUT (AI-generated explanation)

If the mean converged, X_n/n would tend to 0, contradicting the Borel-Cantelli consequence in Statement II.

17. **FORECAST · FP-337** Central Limit Theorem

QUESTION

Twelve independent $U(0,1)$ variables are summed. What is the approximate value of $\mathrm{P}(\mathrm{sum} > 7)$?

OPTIONS

(a) 0.1587

(b) 0.0228

(c) 0.5

(d) 0.3085

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.1587

EXAM SHORTCUT (AI-generated explanation)

> Mean 6 and variance exactly $12/12 = 1$, so $z = 1$ and the tail is 0.1587.

18. **FORECAST · FP-338** Delta Method

QUESTION

X_i are independent $\mathrm{Exp}(\mathrm{rate}\ \lambda)$. Then $\sqrt{n}$ times (the reciprocal of the sample mean minus λ) tends to $N(0, v)$. What is v?

OPTIONS

(a) λ^2

(b) $1/\lambda^2$

(c) λ

(d) λ^4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) λ^2

EXAM SHORTCUT (AI-generated explanation)

> Δ method with $g(x) = 1/x$ at $x = 1/\lambda$: $g'(1/\lambda)^2$ times $1/\lambda^2 = \lambda^4/\lambda^2 = \lambda^2$.

19. FORECAST · FP-339 Weak and Strong Laws

QUESTION

The X_i have unit variance and $\mathrm{Cov}(X_i, X_j) = \rho > 0$ for every pair. What does $\mathrm{Var}(\text{sample mean})$ tend to?

OPTIONS

(a) 0

(b) ρ

(c) 1

(d) $1/n$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) ρ

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{Var}(\bar{X}) = 1/n + (n-1)\rho/n$, which tends to ρ, not 0.

20. FORECAST · FP-340 Central Limit Theorem

QUESTION

For a proportion with a worst-case p, the sample sizes giving $\mathrm{P}(|\hat{p} - p| \leq 0.02)$ at least 0.95 by the central limit theorem and by Chebyshev's inequality are approximately

OPTIONS

(a) 2401 and 12500

(b) 2401 and 2401

(c) 9604 and 12500

(d) 1000 and 5000

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 2401 and 12500

EXAM SHORTCUT (AI-generated explanation)

CLT: $(1.96/0.02)^2(1/4) = 2401$. Chebyshev: $(1/4)/(0.0004 \times 0.05) = 12500$.

21. **FORECAST · FP-341** Central Limit Theorem

QUESTION

Fifty independent exponential variables with mean 2 are summed. What is the approximate value of $\mathrm{P}(\text{sum} > 120)$?

OPTIONS

(a) 0.08

(b) 0.16

(c) 0.02

(d) 0.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.08

EXAM SHORTCUT (AI-generated explanation)

Mean 100 and variance $50(4) = 200$, so $z = 20/14.14 = 1.41$ and the tail is about 0.08.

22. **FORECAST · FP-342** Weak and Strong Laws

QUESTION

X_i are independent and identically distributed with a finite variance. The average of the X_i squared converges almost surely to

OPTIONS

(a) $\mathrm{Var}(X)$

(b) $\mathrm{E}(X^2)$

(c) the square of $\mathrm{E}(X)$

(d) nothing

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\mathrm{E}(X^2)$

EXAM SHORTCUT (AI-generated explanation)

Apply the strong law to the variables X_i^2, whose common mean is $\mathrm{E}(X^2)$.

23. FORECAST · FP-343 Central Limit Theorem

QUESTION

The statistic $\sqrt{n}(\bar{X}-\mu)/S$, with S the sample standard deviation, tends to $N(0,1)$. The justification is

OPTIONS

(a) the central limit theorem alone

(b) the central limit theorem together with Slutsky's theorem, since S tends to σ in probability

(c) the strong law alone

(d) Chebyshev's inequality

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the central limit theorem together with Slutsky's theorem, since S tends to σ in probability

EXAM SHORTCUT (AI-generated explanation)

> The numerator needs the central limit theorem and the random denominator needs Slutsky's theorem.

24. FORECAST · FP-344 Central Limit Theorem

QUESTION

X_i are independent $\text{Bernoulli}(p)$. What does the variance of $\sqrt{n}$(sample mean minus p) tend to?

OPTIONS

(a) 0

(b) $p(1-p)$

(c) 1

(d) $np(1-p)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $p(1-p)$

EXAM SHORTCUT (AI-generated explanation)

> The variance is exactly $p(1-p)$ for every n, so the limit is $p(1-p)$.

25. **FORECAST · FP-373** Weak and Strong Laws

QUESTION

The difference between the weak and the strong law of large numbers is that

OPTIONS

(a) the weak law asserts convergence in probability and the strong law almost sure convergence

(b) the weak law needs a finite variance and the strong law does not

(c) the strong law asserts convergence in distribution

(d) they are logically equivalent

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) the weak law asserts convergence in probability and the strong law almost sure convergence

EXAM SHORTCUT (AI-generated explanation)

> The two laws differ in the MODE of convergence, not in what the limit is.

FORECAST · PROBABILITY

Mathematical and Conditional Expectation

33 forecast problems · 3 concepts

1. FORECAST · FP-046 Expectation and Moments

QUESTION

X follows $N(0,1)$ and $Y=|X|$. What is $\mathrm{E}(Y)$?

OPTIONS

(a) 0

(b) $\sqrt{2/\pi}$

(c) $1/\sqrt{2\pi}$

(d) 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\sqrt{2/\pi}$

EXAM SHORTCUT (AI-generated explanation)

The half-normal mean: $\mathrm{E}|X|=\sigma\sqrt{2/\pi}=\sqrt{2/\pi}$ for the standard normal.

2. FORECAST · FP-061 Tail-Sum Formula

QUESTION

X takes non-negative integer values with $\mathrm{P}(X>x)=1/2^{x+1}$ for $x=0,1,2,\ldots$ What is $\mathrm{E}(X)$?

OPTIONS

(a) $1/2$

(b) 1

(c) 2

(d) $3/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 1

EXAM SHORTCUT (AI-generated explanation)

Tail sum: $\mathrm{E}(X)=\sum\mathrm{P}(X>x)$ = sum of $1/2^{x+1}=1$.

3. FORECAST · FP-062 Conditional Expectation

QUESTION

A miner is at a fork with three doors, chosen with equal probability each time. Door 1 leads to safety in 2 hours, door 2 returns him to the fork after 3 hours and door 3 returns him after 5 hours. What is the expected time to safety?

OPTIONS

(a) $10/3$

(b) 10

(c) 15

(d) 12

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 10

EXAM SHORTCUT (AI-generated explanation)

Condition on the first door: $E = (2+3+5)/3 + (2/3)E$, giving $E = 10$.

4. FORECAST · FP-063 Expectation and Moments

QUESTION

A fair coin is tossed until the first head appears. What is the expected number of tails?

OPTIONS

(a) 1

(b) 2

(c) $1/2$

(d) $3/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 1

EXAM SHORTCUT (AI-generated explanation)

The number of tosses has mean $1/p = 2$, so the number of tails is $q/p = 1$.

5. **FORECAST · FP-064** Tail-Sum Formula

QUESTION

X is a non-negative integer variable with $\mathrm{E}(X) = 3$ and $\mathrm{Var}(X) = 4$. What is the $\sum x\,\mathrm{P}(X > x)$?

OPTIONS

(a) 5

(b) 10

(c) 6

(d) 8

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 5

EXAM SHORTCUT (AI-generated explanation)

The identity gives $(\mathrm{E}(X^2) - \mathrm{E}(X))/2 = (13 - 3)/2 = 5$.

6. **FORECAST · FP-065** Expectation and Moments

QUESTION

If $\mathrm{E}(X) = 2, \mathrm{E}(X^2) = 5$ and $\mathrm{E}(X^3) = 14$, what is the third central moment?

OPTIONS

(a) 0

(b) 2

(c) -2

(d) 4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0

EXAM SHORTCUT (AI-generated explanation)

$\mu_3 = 14 - 3(2)(5) + 2(8) = 14 - 30 + 16 = 0.$

7. **FORECAST · FP-066** Expectation and Moments

QUESTION

X is $U(0,1)$. What is $\mathrm{E}[\min(X,1-X)]$?

OPTIONS

(a) $1/2$

(b) $1/4$

(c) $1/3$

(d) $1/8$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/4$

EXAM SHORTCUT (AI-generated explanation)

> $\min(X,1-X)$ is $U(0,1/2)$, whose mean is $1/4$.

8. **FORECAST · FP-067** Conditional Expectation

QUESTION

Y is $\mathrm{Bernoulli}(1/2)$. Given $Y=0, X$ is $N(0,1)$; given $Y=1, X$ is $N(2,1)$. What is $\mathrm{Var}(X)$?

OPTIONS

(a) 1

(b) 2

(c) 3

(d) 5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 2

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{Var}(X)=\mathrm{E}[\mathrm{Var}(X\mid Y)]+\mathrm{Var}(\mathrm{E}(X\mid Y))=1+\mathrm{Var}(2Y)=1+4(1/4)=2.$

9. **FORECAST · FP-068** Expectation and Moments

QUESTION

What is the $\sum n(1/3)^{n-1}$?

OPTIONS

(a) $3/2$

(b) $9/4$

(c) 3

(d) $9/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $9/4$

EXAM SHORTCUT (AI-generated explanation)

> The series sum of nx^{n-1} is $1/(1-x)^2$; at $x = 1/3$ this is $1/(2/3)^2 = 9/4$.

10. **FORECAST · FP-123** Expectation and Moments

QUESTION

What is the expected number of rolls of a fair die needed to see all six faces at least once?

OPTIONS

(a) 6

(b) 36

(c) 14.7

(d) 12

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 14.7

EXAM SHORTCUT (AI-generated explanation)

> Coupon collector: $6(1 + 1/2 + 1/3 + 1/4 + 1/5 + 1/6) = 6(2.45) = 14.7$.

11. **FORECAST · FP-165** Expectation and Moments

QUESTION

X is $U(0, 1)$. What is $\mathrm{E}[\max(X, 1/2)]$?

OPTIONS

(a) $1/2$

(b) $5/8$

(c) $3/4$

(d) $2/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $5/8$

EXAM SHORTCUT (AI-generated explanation)

> Split at $1/2$: $(1/2)(1/2) + \int_{1/2}^{1} x\,dx = 1/4 + 3/8 = 5/8$.

12. **FORECAST · FP-166** Conditional Expectation

QUESTION

A fair die is rolled until a 6 appears. What is the expected total of all the faces shown, including the final 6?

OPTIONS

(a) 18

(b) 21

(c) 15

(d) 24

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 21

EXAM SHORTCUT (AI-generated explanation)

> The non-six rolls number N - 1 with mean 5 and average face 3, so the total is $15 + 6 = 21$.

13. FORECAST · FP-167 Conditional Expectation

QUESTION

A fair coin is tossed until two consecutive heads appear. What is the expected number of tosses?

OPTIONS

(a) 4

(b) 6

(c) 8

(d) 3

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 6

EXAM SHORTCUT (AI-generated explanation)

The pattern formula $(1+p)/p^2$ gives $(1.5)/(0.25) = 6$.

14. FORECAST · FP-168 Tail-Sum Formula

QUESTION

X is positive with $\mathrm{P}(X > x) = (1 + x)^{-2}$. Which statement is correct?

OPTIONS

(a) the mean is 1 and the variance is 1

(b) the mean is 1 and the variance is infinite

(c) the mean is infinite

(d) the mean is 2 and the variance is infinite

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the mean is 1 and the variance is infinite

EXAM SHORTCUT (AI-generated explanation)

The tail integral gives $\mathrm{E}(X) = 1$, but the second moment integral diverges like the harmonic integral.

15. FORECAST · FP-170 Conditional Expectation

QUESTION

X is $U(-1, 1)$ and $Y = X^2$. What are the best mean-square predictor of Y from X and the best LINEAR predictor?

OPTIONS

(a) X^2 and X

(b) X^2 and $1/3$

(c) $1/3$ and $1/3$

(d) X and X

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) X^2 and $1/3$

EXAM SHORTCUT (AI-generated explanation)

> The best predictor is $\mathrm{E}(Y \mid X) = X^2$; since $\mathrm{Cov}(X, Y) = 0$ the best linear predictor is the constant $\mathrm{E}(Y) = 1/3$.

16. FORECAST · FP-171 Expectation and Moments

QUESTION

$\mathrm{E}(X^2) = 4$ and $\mathrm{E}(Y^2) = 9$. What is the largest possible value of $|\mathrm{E}(XY)|$?

OPTIONS

(a) 36

(b) 6

(c) 13

(d) 12

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 6

EXAM SHORTCUT (AI-generated explanation)

> Cauchy-Schwarz gives the bound $\sqrt{4 \times 9} = 6$ for the absolute value of $\mathrm{E}(XY)$.

17. FORECAST · FP-172 Expectation and Moments

QUESTION

X is $U(1,2)$. Comparing $\mathrm{E}(1/X)$ with $1/\mathrm{E}(X)$:

OPTIONS

(a) they are equal

(b) $\log 2$ is greater than $2/3$

(c) $\log 2$ is less than $2/3$

(d) the comparison is undefined

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\log 2$ is greater than $2/3$

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{E}(1/X) = \log 2 = 0.693$ and $1/\mathrm{E}(X) = 2/3 = 0.667$, and Jensen guarantees the first is larger.

18. FORECAST · FP-173 Expectation and Moments

QUESTION

For $n \geq 2, n$ hats are returned to their owners at random. What is the variance of the number of correct returns?

OPTIONS

(a) 1

(b) $1 - 1/n$

(c) n

(d) $1/n$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 1

EXAM SHORTCUT (AI-generated explanation)

The count has mean 1 and variance 1 for every $n \geq 2$ - a striking exact result.

19. FORECAST · FP-174 Expectation and Moments

QUESTION

X and Y are independent $U(0,1)$. What is $\mathrm{E}|X-Y|$?

OPTIONS

(a) $1/2$

(b) $1/3$

(c) $1/4$

(d) $1/6$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/3$

EXAM SHORTCUT (AI-generated explanation)

The density of $|X-Y|$ is $2(1-u)$ on $(0,1)$, whose mean is $1/3$.

20. FORECAST · FP-175 Expectation and Moments

QUESTION

"Every symmetric distribution has all its odd central moments equal to zero." This statement is

OPTIONS

(a) always true

(b) true only when the relevant moments exist

(c) false for the Laplace distribution

(d) false for the normal distribution

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) true only when the relevant moments exist

EXAM SHORTCUT (AI-generated explanation)

Symmetry gives vanishing odd moments only when they exist; the Cauchy is symmetric with no moments at all.

21. FORECAST · FP-251 Conditional Expectation

QUESTION

Wald's identity $\mathrm{E}(S_N) = \mathrm{E}(N)\,\mathrm{E}(X)$ requires

OPTIONS

(a) only that N is independent of the X_i

(b) that N is a stopping time with a finite mean and that $\mathrm{E}|X|$ is finite

(c) that the X_i are non-negative

(d) that N is bounded

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) that N is a stopping time with a finite mean and that $\mathrm{E}|X|$ is finite

EXAM SHORTCUT (AI-generated explanation)

> Wald holds far beyond independence: N need only be a stopping time with finite mean, with integrable summands.

22. FORECAST · FP-260 Expectation and Moments

QUESTION

Statement I: $\mathrm{E}(X - a)^2$ is minimised at $a = \mathrm{E}(X)$. Statement II: $\mathrm{E}|X - a|$ is minimised at any median of X. Choose the correct code.

OPTIONS

(a) both true, II explains I

(b) both true, II does not explain I

(c) I true, II false

(d) I false, II true

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) both true, II does not explain I

EXAM SHORTCUT (AI-generated explanation)

> Both are true, but they are separate optimality results - one for the mean, one for the median.

23. FORECAST · FP-307 Conditional Expectation

QUESTION

X is geometric on $\{1, 2, \ldots\}$ with success probability p. What is $\mathrm{E}(X \mid X > 1)$?

OPTIONS

(a) $1/p$

(b) $1 + 1/p$

(c) $2/p$

(d) $1/p - 1$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1 + 1/p$

EXAM SHORTCUT (AI-generated explanation)

By memorylessness the excess beyond the first failure is again geometric, so the answer is $1 + 1/p$.

24. FORECAST · FP-308 Expectation and Moments

QUESTION

What is the expected number of draws needed to collect all four of four equally likely coupons?

OPTIONS

(a) 8

(b) $25/3$

(c) 10

(d) 12

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $25/3$

EXAM SHORTCUT (AI-generated explanation)

Coupon collector with $n = 4$: $4(1 + 1/2 + 1/3 + 1/4) = 4(25/12) = 25/3$.

25. FORECAST · FP-309 Expectation and Moments

QUESTION

What is the variance of the sum of n independent fair dice?

OPTIONS

(a) $35n/12$

(b) $35n^2/12$

(c) $91n/12$

(d) 3.5n

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $35n/12$

EXAM SHORTCUT (AI-generated explanation)

A single die has variance $(36-1)/12 = 35/12$, and variances add for independent dice.

26. FORECAST · FP-310 Conditional Expectation

QUESTION

Given Y, the variable X is $\mathrm{Binomial}(Y, 1/2)$, and Y is $\mathrm{Poisson}(4)$. What is $\mathrm{Var}(X)$?

OPTIONS

(a) 1

(b) 2

(c) 4

(d) 3

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 2

EXAM SHORTCUT (AI-generated explanation)

Thinning gives X as $\mathrm{Poisson}(2)$, so the variance is 2.

27. **FORECAST · FP-311** Tail-Sum Formula

QUESTION

X is $\mathrm{Exp}(\lambda)$ and $c > 0$. What is $\mathrm{E}[\min(X, c)]$?

OPTIONS

(a) $1/\lambda - c$

(b) $(1 - e^{-\lambda c})/\lambda$

(c) $ce^{-\lambda c}$

(d) $c/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $(1 - e^{-\lambda c})/\lambda$

EXAM SHORTCUT (AI-generated explanation)

Integrating the survival function up to c gives $(1 - e^{-\lambda c})/\lambda$.

28. **FORECAST · FP-312** Conditional Expectation

QUESTION

A fair gambler starts with 2 units against an opponent with 1 unit, and each play moves one unit either way until someone is ruined. What is the expected number of plays?

OPTIONS

(a) 3

(b) 2

(c) 1

(d) 6

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 2

EXAM SHORTCUT (AI-generated explanation)

The classical duration formula gives $a(N - a) = 2(1) = 2$.

29. FORECAST · FP-313 Conditional Expectation

QUESTION

A fair coin is tossed. What are the expected numbers of tosses until the pattern HT and until the pattern HH first appear?

OPTIONS

(a) 4 and 4

(b) 4 and 6

(c) 6 and 6

(d) 6 and 4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 4 and 6

EXAM SHORTCUT (AI-generated explanation)

HT takes 4 and HH takes 6; the self-overlap of HH costs the extra two tosses.

30. FORECAST · FP-314 Expectation and Moments

QUESTION

A fair die is rolled until a face repeats the face shown immediately before it. What is the expected number of rolls?

OPTIONS

(a) 6

(b) 7

(c) 36

(d) 12

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 7

EXAM SHORTCUT (AI-generated explanation)

After the first roll each further roll matches the previous with probability $1/6$, so the wait is $1+6=7$.

31. **FORECAST · FP-315** Expectation and Moments

QUESTION

What is the expected number of distinct faces seen in 6 rolls of a fair die?

OPTIONS

(a) $6(1-(5/6)^6)$

(b) 3.5

(c) 6

(d) 5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $6(1-(5/6)^6)$

EXAM SHORTCUT (AI-generated explanation)

> Indicators: each face is missed with probability $(5/6)^6$, so the expected count is $6(1-(5/6)^6)=3.99$.

32. **FORECAST · FP-316** Tail-Sum Formula

QUESTION

For a non-negative continuous variable $X, \mathrm{E}(X^2)$ equals

OPTIONS

(a) the $\int x\,\mathrm{P}(X>x)\,dx$

(b) twice the $\int x\,\mathrm{P}(X>x)\,dx$

(c) the $\int \mathrm{P}(X>x)^2\,dx$

(d) the $\int x^2\,\mathrm{P}(X>x)\,dx$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) twice the $\int x\,\mathrm{P}(X>x)\,dx$

EXAM SHORTCUT (AI-generated explanation)

> Integrating by parts gives $\mathrm{E}(X^2)=2$ times the $\int x\,\mathrm{P}(X>x)\,dx$.

33. **FORECAST · FP-367** Expectation and Moments

QUESTION

Which statement about expectation is TRUE in general?

OPTIONS

(a) $\mathrm{E}(XY)$ equals $\mathrm{E}(X)\,\mathrm{E}(Y)$ for all X and Y

(b) $\mathrm{E}(g(X))$ equals $g(\mathrm{E}(X))$ for all g

(c) $\mathrm{E}(X+Y)$ equals $\mathrm{E}(X)+\mathrm{E}(Y)$ whenever both means exist

(d) $\mathrm{E}(1/X)$ equals $1/\,\mathrm{E}(X)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) $\mathrm{E}(X+Y)$ equals $\mathrm{E}(X)+\mathrm{E}(Y)$ whenever both means exist

EXAM SHORTCUT (AI-generated explanation)

> Linearity of expectation needs no independence at all; the other three require special structure.

FORECAST · PROBABILITY

Modes of Convergence and 0-1 Laws

27 forecast problems · 4 concepts

1. FORECAST · FP-077 Almost Sure Convergence

QUESTION

U is $U(0,1)$ and $X_n = n$ times the indicator that $U < 1/n$. Which statement holds?

OPTIONS

(a) X_n converges to 0 almost surely and in mean

(b) X_n converges to 0 almost surely but not in mean

(c) X_n converges to 0 in probability but not almost surely

(d) none of these

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) X_n converges to 0 almost surely but not in mean

EXAM SHORTCUT (AI-generated explanation)

For each fixed $U > 0$ eventually $1/n < U$, so X_n goes to 0 pointwise, yet $\mathrm{E}(X_n) = n(1/n) = 1$ for every n.

2. FORECAST · FP-078 Borel-Cantelli Lemmas

QUESTION

$A_1, A_2, \ldots$ are independent events with $\mathrm{P}(A_n) = 1/n$. What is $\mathrm{P}(A_n \text{ occurs infinitely often})$?

OPTIONS

(a) 0

(b) 1

(c) $1/2$

(d) it cannot be determined

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 1

EXAM SHORTCUT (AI-generated explanation)

The harmonic series diverges and the events are independent, so the second Borel-Cantelli lemma gives probability 1.

3. FORECAST · FP-079 Borel-Cantelli Lemmas

QUESTION

$A_1, A_2, \ldots$ are independent events with $\mathrm{P}(A_n) = 1/n^2$. What is $\mathrm{P}(A_n \text{ occurs infinitely often})$?

OPTIONS

(a) 0

(b) 1

(c) $\pi^2/6$

(d) it cannot be determined

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0

EXAM SHORTCUT (AI-generated explanation)

The sum of $1/n^2$ converges, so the first Borel-Cantelli lemma gives probability 0.

4. FORECAST · FP-080 Convergence in Probability and Distribution

QUESTION

X_n follows $U(-n, n)$. The limiting distribution of X_n is

OPTIONS

(a) uniform on the whole real line

(b) degenerate at 0

(c) non-existent

(d) $N(0, 1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) non-existent

EXAM SHORTCUT (AI-generated explanation)

$F_n(x)$ tends to $1/2$ for every fixed x, and the constant $1/2$ is not a distribution function.

5. **FORECAST · FP-081** Convergence in Probability and Distribution

QUESTION

X_n converges in distribution to the constant c. Then

OPTIONS

(a) X_n converges to c in probability

(b) X_n converges to c almost surely

(c) X_n converges to c in mean square

(d) none of these follows

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) X_n converges to c in probability

EXAM SHORTCUT (AI-generated explanation)

> Convergence in distribution to a CONSTANT is equivalent to convergence in probability to that constant.

6. **FORECAST · FP-082** Kolmogorov Zero-One Law

QUESTION

For a sequence of independent random variables, the event that the series of the X_n converges has probability

OPTIONS

(a) possibly strictly between 0 and 1

(b) exactly 0 or exactly 1

(c) always 1

(d) always 0

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) exactly 0 or exactly 1

EXAM SHORTCUT (AI-generated explanation)

> Convergence of the series is a tail event of an independent sequence, so Kolmogorov's zero-one law forces probability 0 or 1.

7. FORECAST · FP-187 Convergence in Probability and Distribution

QUESTION

X_n is geometric on {1,2,...} with success probability p_n, where np_n tends to λ. Then X_n/n converges in distribution to

OPTIONS

(a) $\mathrm{Poisson}(\lambda)$

(b) an exponential with rate λ

(c) the constant $1/\lambda$

(d) nothing

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) an exponential with rate λ

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{P}(X_n/n > x) = (1 - p_n)^{nx}$, which tends to $e^{-\lambda x}$.

8. FORECAST · FP-188 Almost Sure Convergence

QUESTION

X is $N(0,1)$ and $X_n = (-1)^n X$. In what sense does X_n converge?

OPTIONS

(a) almost surely

(b) in probability

(c) in distribution only

(d) in mean square

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) in distribution only

EXAM SHORTCUT (AI-generated explanation)

Every X_n is $N(0,1)$ so the distributions never change, but the sequence oscillates and never settles.

9. **FORECAST · FP-189** Convergence in Probability and Distribution

QUESTION

X_n converges in distribution to X and Y_n converges in distribution to Y. The product $X_n Y_n$ converges in distribution to XY when

OPTIONS

(a) always

(b) X or Y is a constant

(c) X and Y are independent

(d) never

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) X or Y is a constant

EXAM SHORTCUT (AI-generated explanation)

> Slutsky's theorem needs one of the two limits to be degenerate; otherwise the joint behaviour is not determined.

10. **FORECAST · FP-190** Borel-Cantelli Lemmas

QUESTION

$X_1, X_2, \ldots$ are independent $\mathrm{Exp}(1)$. What are P($X_n > \log n$ infinitely often) and P($X_n > n$ infinitely often)?

OPTIONS

(a) 1 and 1

(b) 0 and 0

(c) 1 and 0

(d) 0 and 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 1 and 0

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{P}(X_n > \log n) = 1/n$ sums to ∞ so the answer is 1; $\mathrm{P}(X_n > n) = e^{-n}$ sums finitely so the answer is 0.

11. **FORECAST · FP-191** Convergence in Probability and Distribution

QUESTION

X_n converges in distribution to X and g is continuous. Then $g(X_n)$ converges to $g(X)$

OPTIONS

(a) in distribution

(b) in probability

(c) almost surely

(d) not necessarily at all

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) in distribution

EXAM SHORTCUT (AI-generated explanation)

The continuous mapping theorem preserves the mode of convergence, so the conclusion is again in distribution.

12. **FORECAST · FP-192** Almost Sure Convergence

QUESTION

Which implication is FALSE in general?

OPTIONS

(a) convergence in mean square implies convergence in mean

(b) convergence in mean implies convergence in probability

(c) almost sure convergence implies convergence in mean

(d) almost sure convergence implies convergence in probability

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) almost sure convergence implies convergence in mean

EXAM SHORTCUT (AI-generated explanation)

n times the indicator that $U < 1/n$ converges almost surely to 0 but has mean 1 throughout.

13. FORECAST · FP-193 Kolmogorov Zero-One Law

QUESTION

$X_1, X_2, \ldots$ are independent $N(0,1)$. What is P(the limit superior of X_n is plus ∞)?

OPTIONS

(a) 0

(b) 1

(c) $1/2$

(d) it depends on the sequence

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 1

EXAM SHORTCUT (AI-generated explanation)

For every M the sum of $\mathrm{P}(X_n > M)$ diverges, so by Borel-Cantelli the sequence exceeds M infinitely often.

14. FORECAST · FP-252 Convergence in Probability and Distribution

QUESTION

Statement I: convergence in probability implies convergence of the expectations. Statement II: convergence in probability implies convergence in distribution. Choose the correct code: (a) both true and II explains I, (b) both true but II does not explain I, (c) I true and II false, (d) I false and II true.

OPTIONS

(a) both true, II explains I

(b) both true, II does not explain I

(c) I true, II false

(d) I false, II true

ANSWER (AI-generated forecast item — no official UPSC key exists)

(d) I false, II true

EXAM SHORTCUT (AI-generated explanation)

Expectations can fail to converge, as n times the indicator that $U < 1/n$ shows; but convergence in probability always implies convergence in distribution.

15. **FORECAST · FP-255** Convergence in Probability and Distribution

QUESTION

Statement I: if X_n and Y_n converge in distribution then $X_n + Y_n$ converges in distribution to the sum of the limits. Statement II: Slutsky's theorem requires one of the two limits to be a constant. Choose the correct code.

OPTIONS

(a) both true, II explains I

(b) both true, II does not explain I

(c) I true, II false

(d) I false, II true

ANSWER (AI-generated forecast item — no official UPSC key exists)

(d) I false, II true

EXAM SHORTCUT (AI-generated explanation)

> Marginal convergence says nothing about the joint law, so the sum need not converge; Slutsky's constant assumption is exactly the repair.

16. **FORECAST · FP-256** Almost Sure Convergence

QUESTION

Statement I: almost sure convergence implies convergence in distribution. Statement II: almost sure convergence implies convergence in probability, which in turn implies convergence in distribution. Choose the correct code.

OPTIONS

(a) both true, II explains I

(b) both true, II does not explain I

(c) I true, II false

(d) I false, II true

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) both true, II explains I

EXAM SHORTCUT (AI-generated explanation)

> The implication chain in Statement II is exactly the proof of Statement I.

17. FORECAST · FP-257 Kolmogorov Zero-One Law

QUESTION

Statement I: for an independent sequence, the event that the limit superior of X_n/n is positive has probability 0 or 1. Statement II: that event is a tail event of an independent sequence. Choose the correct code.

OPTIONS

(a) both true, II explains I

(b) both true, II does not explain I

(c) I true, II false

(d) I false, II true

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) both true, II explains I

EXAM SHORTCUT (AI-generated explanation)

Changing finitely many terms cannot affect the limit superior, so the event is a tail event and Kolmogorov's law applies.

18. FORECAST · FP-261 Convergence in Probability and Distribution

QUESTION

Statement I: if the densities f_n converge pointwise to a density f, then the corresponding variables converge in distribution. Statement II: Scheffe's lemma upgrades pointwise convergence of densities to convergence in total variation. Choose the correct code.

OPTIONS

(a) both true, II explains I

(b) both true, II does not explain I

(c) I true, II false

(d) I false, II true

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) both true, II explains I

EXAM SHORTCUT (AI-generated explanation)

Scheffe's lemma gives convergence in total variation, which is strictly stronger than convergence in distribution.

19. FORECAST · FP-324 Almost Sure Convergence

QUESTION

U is $U(0,1)$ and $X_n = nU^n$. In what sense does X_n converge to 0?

OPTIONS

(a) almost surely and in mean

(b) almost surely but not in mean

(c) in mean but not almost surely

(d) in neither sense

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) almost surely but not in mean

EXAM SHORTCUT (AI-generated explanation)

nu^n tends to 0 for every $u < 1$, but $\mathrm{E}(X_n) = n/(n+1)$ tends to 1.

20. FORECAST · FP-325 Almost Sure Convergence

QUESTION

U is $U(0,1)$ and $X_n = U^n$. In what sense does X_n converge to 0?

OPTIONS

(a) almost surely and in mean

(b) almost surely only

(c) in probability only

(d) in distribution only

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) almost surely and in mean

EXAM SHORTCUT (AI-generated explanation)

u^n tends to 0 pointwise and $\mathrm{E}(U^n) = 1/(n+1)$ tends to 0 as well.

21. **FORECAST · FP-326** Borel-Cantelli Lemmas

QUESTION

X_n are independent with $\mathrm{P}(X_n = 1) = n^{-p}$ and $\mathrm{P}(X_n = 0) = 1 - n^{-p}$. Then X_n tends to 0 almost surely if and only if

OPTIONS

(a) $p > 0$

(b) $p > 1$

(c) $p \geq 1$

(d) $p > 2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $p > 1$

EXAM SHORTCUT (AI-generated explanation)

> The sum of n^{-p} converges exactly when $p > 1$, and Borel-Cantelli then forces only finitely many ones.

22. **FORECAST · FP-327** Convergence in Probability and Distribution

QUESTION

X_n is degenerate at $1/n$, so X_n tends to 0 in distribution, yet $F_n(0) = 0$ while $F(0) = 1$. This is

OPTIONS

(a) a contradiction

(b) allowed, because 0 is a discontinuity point of the limiting distribution function

(c) impossible for degenerate limits

(d) a failure of the central limit theorem

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) allowed, because 0 is a discontinuity point of the limiting distribution function

EXAM SHORTCUT (AI-generated explanation)

> Weak convergence requires $F_n(x)$ to converge to $F(x)$ only at continuity points of F.

23. FORECAST · FP-328 Almost Sure Convergence

QUESTION

The typewriter sequence sets X_n as the indicator that U lies in I_n, where the intervals I_n sweep across $(0, 1)$ with lengths tending to 0. Then X_n tends to 0

OPTIONS

(a) almost surely

(b) in probability and in mean but not almost surely

(c) in distribution only

(d) in mean square but not in probability

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) in probability and in mean but not almost surely

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{P}(X_n = 1)$ is the length of I_n and tends to 0, but every point is covered infinitely often, so no pointwise limit exists.

24. FORECAST · FP-329 Convergence in Probability and Distribution

QUESTION

X_n converges in distribution to X and $|X_n| \le M$ for all n. Then

OPTIONS

(a) $\mathrm{E}(X_n)$ converges to $\mathrm{E}(X)$

(b) $\mathrm{E}(X_n)$ need not converge

(c) X_n converges to X in probability

(d) X_n converges to X almost surely

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\mathrm{E}(X_n)$ converges to $\mathrm{E}(X)$

EXAM SHORTCUT (AI-generated explanation)

Uniform boundedness supplies uniform integrability, so the expectations converge.

25. FORECAST · FP-330 Almost Sure Convergence

QUESTION

If X_n converges to X in probability, then there exists a subsequence converging

OPTIONS

(a) in mean

(b) almost surely

(c) uniformly

(d) in no guaranteed sense

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) almost surely

EXAM SHORTCUT (AI-generated explanation)

Choosing n_k so that $\mathrm{P}(|X_{n_k} - X| > 2^{-k}) < 2^{-k}$ makes Borel-Cantelli apply.

26. FORECAST · FP-368 Convergence in Probability and Distribution

QUESTION

"Almost surely" and "surely" differ because

OPTIONS

(a) an almost sure statement may fail on a set of probability zero

(b) an almost sure statement may fail on a set of positive probability

(c) they are the same thing

(d) almost sure statements involve limits and sure statements do not

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) an almost sure statement may fail on a set of probability zero

EXAM SHORTCUT (AI-generated explanation)

An event of probability 1 may still have a non-empty complement, and that complement is the exceptional set.

27. FORECAST · FP-374 Borel-Cantelli Lemmas

QUESTION

Which pair of statements about the Borel-Cantelli lemmas is correct?

OPTIONS

(a) both lemmas require independence

(b) the first needs no independence; the second requires it

(c) the first requires independence; the second does not

(d) neither requires independence

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the first needs no independence; the second requires it

EXAM SHORTCUT (AI-generated explanation)

> The convergent half is a pure measure-theoretic bound; the divergent half genuinely needs independence.

FORECAST · PROBABILITY

Probability Inequalities

18 forecast problems · 3 concepts

1. FORECAST · FP-083 Chebyshev Inequality

QUESTION

Weekly sales have mean 50 and variance 25. What lower bound does Chebyshev's inequality give for $\mathrm{P}(40 < X < 60)$?

OPTIONS

(a) $3/4$

(b) $1/4$

(c) $8/9$

(d) 0.9544

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $3/4$

EXAM SHORTCUT (AI-generated explanation)

The half-width 10 is $k = 2$ standard deviations, so the bound is $1 - 1/4 = 3/4$.

2. FORECAST · FP-084 Chebyshev Inequality

QUESTION

Chebyshev's inequality gives $\mathrm{P}(|X - \mu| < 3\sigma)$ >=

OPTIONS

(a) $1/9$

(b) $8/9$

(c) $3/4$

(d) 0.9973

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $8/9$

EXAM SHORTCUT (AI-generated explanation)

$1 - 1/9 = 8/9$.

3. **FORECAST · FP-085** Chebyshev Inequality

QUESTION

X has mean 20, and the Chebyshev bound gives $\mathrm{P}(|X - 20| \geq 5) \leq 0.16$. What is $\mathrm{E}(X^2)$?

OPTIONS

(a) 404

(b) 425

(c) 416

(d) 400

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 404

EXAM SHORTCUT (AI-generated explanation)

> $\sigma^2/25 = 0.16$ gives $\sigma^2 = 4$, so $\mathrm{E}(X^2) = 4 + 400 = 404$.

4. **FORECAST · FP-086** Markov Inequality

QUESTION

X is non-negative with $\mathrm{E}(X) = 2$. What upper bound does Markov's inequality give for $\mathrm{P}(X \geq 8)$?

OPTIONS

(a) $1/2$

(b) $1/4$

(c) $1/8$

(d) $1/16$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/4$

EXAM SHORTCUT (AI-generated explanation)

> Markov: $\mathrm{P}(X \geq a) \leq \mathrm{E}(X)/a = 2/8 = 1/4$.

5. **FORECAST · FP-087** Chebyshev Inequality

QUESTION

A $\text{Bernoulli}(p)$ sample is used to estimate p. By Chebyshev's inequality, what is the smallest n guaranteeing $\mathrm{P}(|\hat{p} - p| < 0.1) \geq 0.9$ for every p?

OPTIONS

(a) 100

(b) 250

(c) 1000

(d) 25

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 250

EXAM SHORTCUT (AI-generated explanation)

> $pq \leq 1/4$, so the bound is $1 - 25/n \geq 0.9$, giving $n \geq 250$.

6. **FORECAST · FP-194** One-Sided and Other Inequalities

QUESTION

X has mean 100 and standard deviation 10. What is the sharpest guaranteed bound on $\mathrm{P}(X \geq 120)$?

OPTIONS

(a) $1/4$

(b) $1/5$

(c) $1/8$

(d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/5$

EXAM SHORTCUT (AI-generated explanation)

> Cantelli's one-sided bound $1/(1+k^2)$ with $k = 2$ gives $1/5$, beating Chebyshev's two-sided $1/4$.

7. **FORECAST · FP-195** Chebyshev Inequality

QUESTION

Chebyshev's bound $\mathrm{P}(|X-\mu| \geq k\sigma) \leq 1/k^2$ is attained by

OPTIONS

(a) no distribution

(b) the normal distribution

(c) the three-point law taking -k, 0 and k with probabilities $1/(2k^2), 1-1/k^2$ and $1/(2k^2)$

(d) the uniform distribution

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) the three-point law taking -k, 0 and k with probabilities $1/(2k^2), 1-1/k^2$ and $1/(2k^2)$

EXAM SHORTCUT (AI-generated explanation)

> That three-point law has mean 0, variance 1 and puts exactly $1/k^2$ outside $k\sigma$.

8. **FORECAST · FP-196** Markov Inequality

QUESTION

X is non-negative with $\mathrm{E}(X^2)=4$. What is the best Markov-type bound on $\mathrm{P}(X \geq 4)$?

OPTIONS

(a) $1/2$

(b) $1/4$

(c) $1/16$

(d) 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/4$

EXAM SHORTCUT (AI-generated explanation)

> Apply Markov to X^2: $\mathrm{P}(X^2 \geq 16) \leq 4/16 = 1/4$.

9. FORECAST · FP-197 Chebyshev Inequality

QUESTION

$X_1, \ldots, X_{100}$ are independent with mean 0 and variance 1. What bound does Chebyshev give for $\mathrm{P}(|\sum| \geq 50)$?

OPTIONS

(a) $1/25$

(b) $1/4$

(c) $1/2$

(d) $1/50$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/25$

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{Var}(\mathrm{sum}) = 100$, so the bound is $100/2500 = 1/25$.

10. FORECAST · FP-198 Markov Inequality

QUESTION

X is non-negative with $\mathrm{E}(X) = 5$. The Markov bound on $\mathrm{P}(X \geq 3)$ is

OPTIONS

(a) $3/5$

(b) $5/3$, which carries no information

(c) 0.6

(d) $2/5$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $5/3$, which carries no information

EXAM SHORTCUT (AI-generated explanation)

> Markov gives $5/3$, which exceeds 1 and is therefore vacuous.

11. **FORECAST · FP-199** One-Sided and Other Inequalities

QUESTION

As a function of $r > 0$, the quantity $(\mathrm{E}|X|^r)^{1/r}$ is

OPTIONS

(a) decreasing

(b) non-decreasing

(c) constant

(d) unimodal

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) non-decreasing

EXAM SHORTCUT (AI-generated explanation)

Lyapunov's inequality says the norms increase with the order r.

12. **FORECAST · FP-331** Chebyshev Inequality

QUESTION

X is $\mathrm{Poisson}(\lambda)$. What bound does Chebyshev's inequality give for $\mathrm{P}(|X-\lambda| \geq \lambda)$?

OPTIONS

(a) $1/\lambda$

(b) $1/\lambda^2$

(c) λ

(d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/\lambda$

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{Var} = \lambda$ and the deviation is λ, so the bound is $\lambda/\lambda^2 = 1/\lambda$.

13. FORECAST · FP-332 Chebyshev Inequality

QUESTION

The population standard deviation is 2. By Chebyshev's inequality, what is the smallest n making P(|sample mean - μ| < 0.5) at least 0.95?

OPTIONS

(a) 64

(b) 320

(c) 80

(d) 160

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 320

EXAM SHORTCUT (AI-generated explanation)

$4/n$ divided by 0.25 must not exceed 0.05, giving $n \geq 320$.

14. FORECAST · FP-333 One-Sided and Other Inequalities

QUESTION

X takes values in $[0, 1]$ with mean μ. Which bound is valid for every such X?

OPTIONS

(a) $\mathrm{Var}(X) \leq \mu(1-\mu)$

(b) $\mathrm{Var}(X) \leq \mu^2$

(c) $\mathrm{P}(X \geq 1/2) \geq 2\mu$

(d) $\mathrm{Var}(X) \geq \mu(1-\mu)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\mathrm{Var}(X) \leq \mu(1-\mu)$

EXAM SHORTCUT (AI-generated explanation)

$X^2 \leq X$ on $[0, 1]$ gives $\mathrm{E}(X^2) \leq \mu$, hence $\mathrm{Var} \leq \mu - \mu^2$.

15. **FORECAST · FP-334** Chebyshev Inequality

QUESTION

X is $\mathrm{Binomial}(100, 1/2)$. What bound does Chebyshev's inequality give for $\mathrm{P}(|X - 50| \geq 10)$?

OPTIONS

(a) $1/4$

(b) $1/10$

(c) $1/100$

(d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/4$

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{Var} = 25$, so the bound is $25/100 = 1/4$, while the exact probability is only about 0.057.

16. **FORECAST · FP-335** One-Sided and Other Inequalities

QUESTION

The bound $\mathrm{P}(X \geq a) \leq e^{-ta} M_X(t)$, valid for every $t > 0$, is

OPTIONS

(a) Chebyshev's inequality

(b) Markov's inequality applied to e^{tX}, the Chernoff bound

(c) Jensen's inequality

(d) Cantelli's inequality

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) Markov's inequality applied to e^{tX}, the Chernoff bound

EXAM SHORTCUT (AI-generated explanation)

Applying Markov to the positive variable e^{tX} and then optimising over t is the Chernoff device.

17. **FORECAST · FP-336** One-Sided and Other Inequalities

QUESTION

The one-sided lower-tail bound $\mathrm{P}(X - \mu \le -k\sigma)$ is at most

OPTIONS

(a) $1/k^2$

(b) $1/(1+k^2)$

(c) $1/(2k^2)$

(d) $1/k$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/(1+k^2)$

EXAM SHORTCUT (AI-generated explanation)

> Cantelli's inequality applied to -X gives $1/(1+k^2)$.

18. **FORECAST · FP-369** One-Sided and Other Inequalities

QUESTION

Jensen's inequality states that for a convex g,

OPTIONS

(a) $\mathrm{E}(g(X)) \le g(\mathrm{E}(X))$

(b) $\mathrm{E}(g(X)) \ge g(\mathrm{E}(X))$

(c) $\mathrm{E}(g(X)) = g(\mathrm{E}(X))$

(d) no general relation holds

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\mathrm{E}(g(X)) \ge g(\mathrm{E}(X))$

EXAM SHORTCUT (AI-generated explanation)

> A convex function lies above its tangent at the mean, so averaging can only raise the value.

FORECAST · PROBABILITY

Random Vectors and Joint Distributions

84 forecast problems · 5 concepts

1. FORECAST · FP-036 Joint and Marginal Distributions

QUESTION

The joint density is $f(x,y) = k(x+y)$ on the unit square. What is k?

OPTIONS

(a) 1

(b) 2

(c) $1/2$

(d) 4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 1

EXAM SHORTCUT (AI-generated explanation)

The $\int x+y$ over the unit square is $1/2+1/2=1$, so $k=1$.

2. FORECAST · FP-037 Independence and Covariance

QUESTION

For $f(x,y) = x+y$ on the unit square, what is $\mathrm{Cov}(X,Y)$?

OPTIONS

(a) 0

(b) $1/144$

(c) $-1/144$

(d) $-1/12$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) $-1/144$

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{E}(XY) = 1/3$ and $\mathrm{E}(X) = \mathrm{E}(Y) = 7/12$, so $\mathrm{Cov} = 1/3 - 49/144 = -1/144$.

3. FORECAST · FP-038 Conditional Distributions

QUESTION

The joint density is $f(x, y) = 8xy$ on $0 < x < y < 1$. What is $\mathrm{E}(X \mid Y = y)$?

OPTIONS

(a) $y/2$

(b) $2y/3$

(c) y

(d) $y^2/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $2y/3$

EXAM SHORTCUT (AI-generated explanation)

Given $Y = y$, the conditional density is $2x/y^2$ on $(0, y)$, which is $\mathrm{Beta}(2, 1)$ scaled to $(0, y)$: mean $2y/3$.

4. FORECAST · FP-039 Joint and Marginal Distributions

QUESTION

The joint density is $f(x, y) = 2$ on $0 < x < y < 1$. What is $\mathrm{P}(X + Y < 1)$?

OPTIONS

(a) $1/4$

(b) $1/2$

(c) $1/3$

(d) $3/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/2$

EXAM SHORTCUT (AI-generated explanation)

The region is the triangle with vertices $(0, 0), (0, 1), (1/2, 1/2)$, of area $1/4$; times the density 2 gives $1/2$.

5. FORECAST · FP-040 Joint and Marginal Distributions

QUESTION

For $f(x,y) = 2$ on $0 < x < y < 1$, the marginal density of X is

OPTIONS

(a) 2x

(b) $2(1-x)$

(c) 1

(d) x

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $2(1-x)$

EXAM SHORTCUT (AI-generated explanation)

Integrate y from x to 1: $2(1-x)$.

6. FORECAST · FP-041 Functions of Random Variables

QUESTION

X and Y are independent $\mathrm{Exp}(1)$ variables. With $U = X + Y$ and $V = X/(X+Y)$, which statement is correct?

OPTIONS

(a) V is $U(0,1)$ and independent of U

(b) V follows $\mathrm{Beta}(2,1)$

(c) U and V are dependent

(d) U follows $\mathrm{Exp}(2)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) V is $U(0,1)$ and independent of U

EXAM SHORTCUT (AI-generated explanation)

$\Gamma(1,1)$ over $\Gamma(1,1)$ gives $V \sim \mathrm{Beta}(1,1) = U(0,1)$, independent of the total $U \sim \Gamma(2,1)$.

7. FORECAST · FP-042 Functions of Random Variables

QUESTION

X and Y are independent $U(0,1)$. What is $\mathrm{P}(XY < 1/2)$?

OPTIONS

(a) $1/2$

(b) $(1+\log 2)/2$

(c) $1-\log 2$

(d) $\log 2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $(1+\log 2)/2$

EXAM SHORTCUT (AI-generated explanation)

Area below the hyperbola: $1/2+\int_{1/2}^{1} 1/(2x)\,dx = 1/2+(\log 2)/2$.

8. FORECAST · FP-043 Independence and Covariance

QUESTION

X is $U(-1,1)$ and $Y = X^2$. Consider Statement I: $\mathrm{Cov}(X,Y)=0$. Statement II: X and Y are independent. Which is correct?

OPTIONS

(a) Both statements are true

(b) I is true, II is false

(c) I is false, II is true

(d) Both statements are false

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) I is true, II is false

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{E}(X^3)=0$ by symmetry so the covariance vanishes, yet Y is an exact function of X, so they cannot be independent.

9. FORECAST · FP-044 Conditional Distributions

QUESTION

The joint pmf is $p(x, y) = (x + y)/21$ for x in {1,2,3} and y in {1,2}. What is $\mathrm{P}(X = 2 \mid Y = 1)$?

OPTIONS

(a) $3/21$

(b) $1/3$

(c) $3/12$

(d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/3$

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{P}(Y = 1) = (2 + 3 + 4)/21 = 9/21$ and $\mathrm{P}(X = 2, Y = 1) = 3/21$, so the ratio is $3/9 = 1/3$.

10. FORECAST · FP-045 Joint and Marginal Distributions

QUESTION

For the joint pmf $p(x, y) = (x + y)/21$ on x in {1,2,3}, y in {1,2}, what is $\mathrm{P}(Y = 2)$?

OPTIONS

(a) $3/7$

(b) $4/7$

(c) $1/2$

(d) $12/21$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $4/7$

EXAM SHORTCUT (AI-generated explanation)

Sum the $y = 2$ row: $(3 + 4 + 5)/21 = 12/21 = 4/7$.

11. **FORECAST · FP-047** Functions of Random Variables

QUESTION

X and Y are independent $\mathrm{Exp}(1)$. Then X - Y follows

OPTIONS

(a) $\mathrm{Exp}(1)$

(b) $\mathrm{Laplace}(0, 1)$

(c) $N(0, 2)$

(d) Cauchy

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\mathrm{Laplace}(0, 1)$

EXAM SHORTCUT (AI-generated explanation)

> The difference of two independent exponentials with the same rate is the standard Laplace.

12. **FORECAST · FP-048** Conditional Distributions

QUESTION

The joint density is $f(x, y) = e^{-y}$ on $0 < x < y < \infty$. What is $\mathrm{E}(X)$?

OPTIONS

(a) $1/2$

(b) 1

(c) 2

(d) e^{-1}

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 1

EXAM SHORTCUT (AI-generated explanation)

> X given Y is $U(0, Y)$ and Y is $\Gamma(2, 1)$ with mean 2, so $\mathrm{E}(X) = \mathrm{E}(Y)/2 = 1$.

13. **FORECAST · FP-049** Joint and Marginal Distributions

QUESTION

For $f(x, y) = e^{-y}$ on $0 < x < y < \infty$, the marginal density of X is

OPTIONS

(a) e^{-x}

(b) xe^{-x}

(c) e^{-2x}

(d) $2e^{-2x}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) e^{-x}

EXAM SHORTCUT (AI-generated explanation)

Integrate y from x to ∞: e^{-x}.

14. **FORECAST · FP-050** Compound and Mixture Distributions

QUESTION

The joint density is $f(x, y) = (2\pi y)^{-1/2} \exp(-(x-y)^2/(2y))e^{-y}$ for x real and $y > 0$. What is $\mathrm{Var}(X)$?

OPTIONS

(a) 1

(b) 2

(c) 3

(d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 2

EXAM SHORTCUT (AI-generated explanation)

Hidden hierarchy: X given Y is $N(Y, Y)$ and Y is $\mathrm{Exp}(1)$, so $\mathrm{Var}(X) = \mathrm{E}(Y) + \mathrm{Var}(Y) = 1 + 1 = 2$.

15. FORECAST · FP-051 Independence and Covariance

QUESTION

X and Y are independent $N(0,1)$. With $U = X + Y$ and $V = X - Y$, which statement is correct?

OPTIONS

(a) U and V are uncorrelated but dependent

(b) U and V are independent

(c) $\mathrm{Cov}(U, V) = 2$

(d) V follows $N(0,1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) U and V are independent

EXAM SHORTCUT (AI-generated explanation)

(U,V) is jointly normal and $\mathrm{Cov}(U, V) = \mathrm{Var}(X) - \mathrm{Var}(Y) = 0$, and for a joint normal zero covariance means independence.

16. FORECAST · FP-052 Functions of Random Variables

QUESTION

(X, Y, Z) has joint density f. If $U = 2X, V = 2Y$ and $W = 2Z$, the joint density of (U, V, W) is

OPTIONS

(a) $8f(u/2, v/2, w/2)$

(b) $f(u/2, v/2, w/2)/8$

(c) $f(2u, 2v, 2w)/8$

(d) $2f(u/2, v/2, w/2)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $f(u/2, v/2, w/2)/8$

EXAM SHORTCUT (AI-generated explanation)

The Jacobian of the inverse map is $(1/2)^3 = 1/8$, so the new density is f at the halved arguments divided by 8.

17. **FORECAST · FP-053** Joint and Marginal Distributions

QUESTION

The joint density is $f(x,y) = 6x$ on $0 < x < y < 1$. The marginal distribution of X is

OPTIONS

(a) $\mathrm{Beta}(2,2)$

(b) the density $3x^2$

(c) $U(0,1)$

(d) $\mathrm{Beta}(1,2)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\mathrm{Beta}(2,2)$

EXAM SHORTCUT (AI-generated explanation)

> Integrating y from x to 1 gives $6 \times (1-x)$, which is the $\mathrm{Beta}(2,2)$ density.

18. **FORECAST · FP-054** Joint and Marginal Distributions

QUESTION

For $f(x,y) = 6x$ on $0 < x < y < 1$, what is $\mathrm{P}(X > 1/2)$?

OPTIONS

(a) $1/4$

(b) $1/2$

(c) $3/4$

(d) $1/8$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/2$

EXAM SHORTCUT (AI-generated explanation)

> X is $\mathrm{Beta}(2,2)$, which is symmetric about $1/2$, so $\mathrm{P}(X > 1/2) = 1/2$.

19. **FORECAST · FP-055** Compound and Mixture Distributions

QUESTION

X given $Y = y$ is $\text{Poisson}(y)$ and Y is $\text{Exp}(1)$. What is the marginal $\text{P}(X = k)$?

OPTIONS

(a) $1/2^{k+1}$

(b) $e^{-1}/k!$

(c) $1/(k+1)!$

(d) $(1/2)^k$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/2^{k+1}$

EXAM SHORTCUT (AI-generated explanation)

Integrating $e^{-y}y^k/k!$ times e^{-y} gives $k!/(k!2^{k+1}) = 1/2^{k+1}$: geometric on {0,1,...} with $p = 1/2$.

20. **FORECAST · FP-056** Compound and Mixture Distributions

QUESTION

X given $Y = y$ is $\text{Poisson}(y)$ and Y is $\text{Exp}(1)$. What is $\text{Var}(X)$?

OPTIONS

(a) 1

(b) 2

(c) 3

(d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 2

EXAM SHORTCUT (AI-generated explanation)

$\text{Var}(X) = \text{E}[\text{Var}(X \mid Y)] + \text{Var}(\text{E}(X \mid Y)) = \text{E}(Y) + \text{Var}(Y) = 1 + 1 = 2.$

21. **FORECAST · FP-057** Functions of Random Variables

QUESTION

X and Y are independent $U(0,1)$. What is the density of $Z = X + Y$ at $z = 1/2$?

OPTIONS

(a) $1/2$

(b) 1

(c) $1/4$

(d) $3/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/2$

EXAM SHORTCUT (AI-generated explanation)

> The sum of two independent uniforms is triangular with density z on $(0,1)$, so $f(1/2) = 1/2$.

22. **FORECAST · FP-059** Functions of Random Variables

QUESTION

X and Y are independent with X following $\mathrm{Exp}(\lambda)$ and Y following $\mathrm{Exp}(\mu)$. What is $\mathrm{P}(X < Y)$?

OPTIONS

(a) $\lambda/(\lambda + \mu)$

(b) $\mu/(\lambda + \mu)$

(c) $1/2$

(d) $\lambda\mu/(\lambda + \mu)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\lambda/(\lambda + \mu)$

EXAM SHORTCUT (AI-generated explanation)

> The race between two exponentials is won by the faster rate: $\lambda/(\lambda + \mu)$.

23. FORECAST · FP-060 Joint and Marginal Distributions

QUESTION

The joint density is the constant c on the triangle $x > 0, y > 0, x + y < 2$. What are c and $\mathrm{E}(X)$?

OPTIONS

(a) $1/2$ and $2/3$

(b) $1/2$ and 1

(c) $1/4$ and 1

(d) 1 and $2/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/2$ and $2/3$

EXAM SHORTCUT (AI-generated explanation)

The triangle has area 2, so $c = 1/2$; the centroid of a right triangle with legs 2 gives $\mathrm{E}(X) = 2/3$.

24. FORECAST · FP-107 Joint and Marginal Distributions

QUESTION

X and Y are independent $U(0, 1)$. What is P($|X - Y| < 1/4$ and $X + Y > 1$)?

OPTIONS

(a) $7/16$

(b) $7/32$

(c) $1/8$

(d) $3/16$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $7/32$

EXAM SHORTCUT (AI-generated explanation)

The band $|X - Y| < 1/4$ has area $1 - (3/4)^2 = 7/16$, and the map (x,y) to $(1 - x, 1 - y)$ splits it in half.

25. **FORECAST · FP-133** Functions of Random Variables

QUESTION

X and Y are independent $\mathrm{Exp}(\lambda)$. What is $\mathrm{P}(X > 2Y)$?

OPTIONS

(a) $1/2$

(b) $1/3$

(c) $1/4$

(d) $2/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/3$

EXAM SHORTCUT (AI-generated explanation)

Condition on Y: the $\int e^{-2\lambda y} \lambda e^{-\lambda y}\, dy = 1/3$.

26. **FORECAST · FP-139** Functions of Random Variables

QUESTION

X and Y are independent Γ variables with shapes a_1 and a_2 and a common rate. Then X/Y follows

OPTIONS

(a) Beta of the first kind with parameters a_1 and a_2

(b) Beta of the second kind with parameters a_1 and a_2

(c) a Γ distribution with shape $a_1 - a_2$

(d) a Cauchy distribution

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) Beta of the second kind with parameters a_1 and a_2

EXAM SHORTCUT (AI-generated explanation)

The ratio of independent Gammas with a common scale is Beta of the second kind, with mean $a_1/(a_2 - 1)$.

27. **FORECAST · FP-140** Functions of Random Variables

QUESTION

X and Y are independent $\mathrm{Exp}(1)$. What is $\mathrm{P}(X > 3Y)$?

OPTIONS

(a) $1/3$

(b) $1/4$

(c) $1/2$

(d) e^{-3}

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/4$

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{P}(X > cY) = 1/(1 + c) = 1/4$ for $c = 3$.

28. **FORECAST · FP-141** Functions of Random Variables

QUESTION

X and Y are independent $N(0, 1)$. What is $\mathrm{P}(X^2 + Y^2 \leq 2\log 2)$?

OPTIONS

(a) $1/4$

(b) $1/2$

(c) $\log 2$

(d) $1 - 1/e$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/2$

EXAM SHORTCUT (AI-generated explanation)

$X^2 + Y^2$ is exponential with mean 2, so the probability is $1 - e^{-\log 2} = 1/2$.

29. FORECAST · FP-142 Conditional Distributions

QUESTION

X is $U(0,1)$ and, given $X = x, Y$ is $U(0,x)$. What is the marginal density of Y on $(0,1)$?

OPTIONS

(a) 1

(b) $-\log y$

(c) $2(1-y)$

(d) $1/y$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $-\log y$

EXAM SHORTCUT (AI-generated explanation)

Integrating $1/x$ from y to 1 gives $-\log y$.

30. FORECAST · FP-143 Conditional Distributions

QUESTION

X is $U(0,1)$ and, given $X = x, Y$ is $U(0,x)$. What is $\mathrm{E}(X \mid Y = y)$?

OPTIONS

(a) $(1+y)/2$

(b) $(1-y)/(-\log y)$

(c) y

(d) $2y/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $(1-y)/(-\log y)$

EXAM SHORTCUT (AI-generated explanation)

Divide $\int_y^1 x(1/x)\,dx = 1-y$ by the marginal $-\log y$.

31. **FORECAST · FP-144** Functions of Random Variables

QUESTION

X and Y are independent $U(0, 1)$, with $U = X + Y$ and $V = X - Y$. What is the joint density of (U, V) on its support?

OPTIONS

(a) 1

(b) 2

(c) $1/2$

(d) u

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) $1/2$

EXAM SHORTCUT (AI-generated explanation)

> The Jacobian of the inverse map is $1/2$, so the density is the constant $1/2$ on the rotated square.

32. **FORECAST · FP-145** Independence and Covariance

QUESTION

The joint density is $f(x, y) = cxy$ on $0 < y < x < 1$. What is c, and are X and Y independent?

OPTIONS

(a) 4 and yes

(b) 8 and no

(c) 8 and yes

(d) 4 and no

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 8 and no

EXAM SHORTCUT (AI-generated explanation)

> The $\int xy$ over the triangle is $1/8$, so $c = 8$; the support is not a rectangle, so they are dependent.

33. **FORECAST · FP-146** Conditional Distributions

QUESTION

X and Y are independent $\mathrm{Exp}(1)$. What is $\mathrm{Var}(X \mid X+Y=s)$?

OPTIONS

(a) $s/2$

(b) $s^2/12$

(c) $s^2/4$

(d) 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $s^2/12$

EXAM SHORTCUT (AI-generated explanation)

Given the total s, X is $U(0,s)$, whose variance is $s^2/12$.

34. **FORECAST · FP-147** Independence and Covariance

QUESTION

X is $\mathrm{Binomial}(2, 1/2)$ and $Y = |X-1|$. Which statement is correct?

OPTIONS

(a) $\mathrm{Cov}(X,Y) = 0$ and they are independent

(b) $\mathrm{Cov}(X,Y) = 0$ but they are dependent

(c) $\mathrm{Cov}(X,Y) = 1/4$

(d) $\mathrm{Cov}(X,Y) = -1/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\mathrm{Cov}(X,Y) = 0$ but they are dependent

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{E}(XY) = 1/2 = \mathrm{E}(X)\,\mathrm{E}(Y)$, so the covariance vanishes, yet Y is a function of X.

35. FORECAST · FP-148 Functions of Random Variables

QUESTION

X is $\mathrm{Exp}(1)$. The integer part of X follows

OPTIONS

(a) $\mathrm{Poisson}(1)$

(b) a geometric distribution on {0,1,...} with $p = 1 - e^{-1}$

(c) a geometric distribution with $p = e^{-1}$

(d) no standard distribution

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) a geometric distribution on {0,1,...} with $p = 1 - e^{-1}$

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{P}(\mathrm{floor}(X) = k) = e^{-k} - e^{-(k+1)} = e^{-k}(1 - e^{-1}), a$ geometric with failure probability e^{-1}.

36. FORECAST · FP-149 Functions of Random Variables

QUESTION

X is uniform on $(-\pi/2, \pi/2)$. Then $\tan X$ follows

OPTIONS

(a) $N(0,1)$

(b) a Laplace distribution

(c) the standard Cauchy distribution

(d) the uniform distribution on the whole line

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) the standard Cauchy distribution

EXAM SHORTCUT (AI-generated explanation)

A uniform angle through the tangent gives exactly the standard Cauchy.

37. **FORECAST · FP-150** Joint and Marginal Distributions

QUESTION

The joint density is $f(x, y) = c(x + y)e^{-(x+y)}$ for $x > 0$ and $y > 0$. What is c?

OPTIONS

(a) 1

(b) $1/2$

(c) 2

(d) $1/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/2$

EXAM SHORTCUT (AI-generated explanation)

The double $\int (x + y)e^{-(x+y)}$ is 2, so $c = 1/2$.

38. **FORECAST · FP-151** Independence and Covariance

QUESTION

For $f(x, y) = (1/2)(x + y)e^{-(x+y)}$ on the positive quadrant, what is $\mathrm{Cov}(X, Y)$?

OPTIONS

(a) 0

(b) $-1/4$

(c) $1/4$

(d) $-1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $-1/4$

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{E}(XY) = 2$ and $\mathrm{E}(X) = \mathrm{E}(Y) = 3/2$, so $\mathrm{Cov} = 2 - 9/4 = -1/4$.

39. FORECAST · FP-152 Functions of Random Variables

QUESTION

X is $U(0,1)$ and $Y = X(1-X)$. What is $\mathrm{P}(Y \leq 3/16)$?

OPTIONS

(a) $3/16$

(b) $1/4$

(c) $1/2$

(d) $3/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) $1/2$

EXAM SHORTCUT (AI-generated explanation)

$x(1-x) \leq 3/16$ means $(4x-1)(4x-3) \geq 0$, i.e. $x \leq 1/4$ or $x \geq 3/4$, of total length $1/2$.

40. FORECAST · FP-153 Functions of Random Variables

QUESTION

X and Y are independent $N(0,1)$. What is $\mathrm{P}(X > |Y|)$?

OPTIONS

(a) $1/2$

(b) $1/4$

(c) $1/8$

(d) $1/\pi$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/4$

EXAM SHORTCUT (AI-generated explanation)

The pair is rotationally symmetric, and $X > |Y|$ is a sector of angle $\pi/2$ out of 2π.

41. **FORECAST · FP-154** Conditional Distributions

QUESTION

X is $\mathrm{Poisson}(2)$ and Y is $\mathrm{Poisson}(3)$, independent. What is $\mathrm{E}(X \mid X+Y=10)$?

OPTIONS

(a) 5

(b) 4

(c) 2

(d) 6

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 4

EXAM SHORTCUT (AI-generated explanation)

Given the total, X is $\mathrm{Binomial}(10, 2/5)$, whose mean is 4.

42. **FORECAST · FP-155** Compound and Mixture Distributions

QUESTION

X is $\mathrm{Binomial}(n, p)$ and, given $X = x, Y$ is $\mathrm{Binomial}(x, r)$. The marginal distribution of Y is

OPTIONS

(a) $\mathrm{Binomial}(n, pr)$

(b) $\mathrm{Binomial}(n, p + r - pr)$

(c) hypergeometric

(d) Poisson

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\mathrm{Binomial}(n, pr)$

EXAM SHORTCUT (AI-generated explanation)

Two-stage thinning multiplies the probabilities: each of the n trials contributes to Y with probability pr.

43. **FORECAST · FP-156** Functions of Random Variables

QUESTION

X is $U(0,1)$ and Y is $\mathrm{Exp}(1)$, independent. What is the density of $Z = X + Y$ at $z = 1$?

OPTIONS

(a) e^{-1}

(b) $1 - e^{-1}$

(c) 1

(d) e - 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1 - e^{-1}$

EXAM SHORTCUT (AI-generated explanation)

For $0 < z < 1$ the convolution is $1 - e^{-z}$; at $z = 1$ this is $1 - e^{-1}$.

44. **FORECAST · FP-157** Independence and Covariance

QUESTION

$\mathrm{Var}(X) = \mathrm{Var}(Y) = 4$ and $\mathrm{Corr}(X, Y) = 0.5$. What is $\mathrm{Cov}(X + Y, X - Y)$?

OPTIONS

(a) 2

(b) 0

(c) 4

(d) -2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{Cov}(X + Y, X - Y) = \mathrm{Var}(X) - \mathrm{Var}(Y) = 0$, whatever the correlation is.

45. FORECAST · FP-158 Independence and Covariance

QUESTION

X is $U(0,1)$. What is $\mathrm{Corr}(X, X^2)$?

OPTIONS

(a) 0

(b) 1

(c) $\sqrt{15}/4$

(d) $1/\sqrt{12}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) $\sqrt{15}/4$

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{Cov} = 1/12, \mathrm{Var}(X) = 1/12$ and $\mathrm{Var}(X^2) = 4/45$, giving a correlation of $\sqrt{15}/4 = 0.968$.

46. FORECAST · FP-159 Joint and Marginal Distributions

QUESTION

The joint density is $f(x, y) = 1/x$ on $0 < y < x < 1$. What is $\mathrm{P}(Y > 1/2)$?

OPTIONS

(a) $1/2$

(b) $(1 - \log 2)/2$

(c) $\log 2$

(d) $1/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $(1 - \log 2)/2$

EXAM SHORTCUT (AI-generated explanation)

> The marginal of Y is $-\log y$, and integrating it from $1/2$ to 1 gives $(1 - \log 2)/2$.

47. **FORECAST · FP-160** Independence and Covariance

QUESTION

X and Y are independent, each with variance 1. With $U = X\cos t + Y\sin t$ and $V = -X\sin t + Y\cos t$, which holds for every t?

OPTIONS

(a) U and V are independent

(b) U and V are uncorrelated

(c) $\mathrm{Var}(U) = \cos^2 t$

(d) $\mathrm{Cov}(U, V) = \sin 2t$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) U and V are uncorrelated

EXAM SHORTCUT (AI-generated explanation)

> The rotation makes the covariance $-\cos t\sin t + \sin t\cos t = 0$, but independence needs normality.

48. **FORECAST · FP-161** Independence and Covariance

QUESTION

X_1, X_2 and X_3 have unit variances and a common pairwise correlation ρ. What is the feasible range of ρ?

OPTIONS

(a) -1 to 1

(b) $-1/2$ to 1

(c) 0 to 1

(d) $-1/3$ to 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $-1/2$ to 1

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{Var}(X_1 + X_2 + X_3) = 3 + 6\rho$ must be non-negative, so $\rho \geq -1/2$.

49. **FORECAST · FP-162** Independence and Covariance

QUESTION

X_1, X_2, X_3 have unit variances and common pairwise correlation ρ. What is $\mathrm{Corr}(X_1, X_1 + X_2 + X_3)$?

OPTIONS

(a) $1/\sqrt{3}$

(b) $\sqrt{(1+2\rho)/3}$

(c) $(1+2\rho)/3$

(d) $1/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\sqrt{(1+2\rho)/3}$

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{Cov} = 1 + 2\rho$ and $\mathrm{Var(sum)} = 3 + 6\rho$, so the correlation is $\sqrt{(1+2\rho)/3}$.

50. **FORECAST · FP-163** Conditional Distributions

QUESTION

X and Y are independent $U(0,1)$. What is P($Y > X^2$ given $Y > X$)?

OPTIONS

(a) $2/3$

(b) $1/2$

(c) 1

(d) $5/6$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 1

EXAM SHORTCUT (AI-generated explanation)

> On $(0,1)$ we have $x^2 \le x$, so $Y > X$ already implies $Y > X^2$ and the conditional probability is 1.

51. FORECAST · FP-164 Independence and Covariance

QUESTION

X and Y each take values 0 and 1 with $\mathrm{P}(X = 1) = \mathrm{P}(Y = 1) = 1/2$ and $\mathrm{P}(X = Y = 1) = a$. What is $\mathrm{Corr}(X, Y)$?

OPTIONS

(a) 4a - 1

(b) 2a - 1

(c) $a - 1/4$

(d) 4a

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 4a - 1

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{Cov} = a - 1/4$ and each variance is $1/4$, so the correlation is 4a - 1.

52. FORECAST · FP-219 Independence and Covariance

QUESTION

X and Y are independent $\mathrm{Exp}(1)$. Which pair of statements about $U = X + Y$ and $V = X/Y$ is correct?

OPTIONS

(a) U and V are independent

(b) U and V are dependent with $\mathrm{Corr}(U, V) = 1/2$

(c) V is $U(0, 1)$

(d) U is $\mathrm{Exp}(2)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) U and V are independent

EXAM SHORTCUT (AI-generated explanation)

> X/Y is a function of $X/(X + Y)$, which is independent of the total under the Gamma-Beta factorisation.

53. **FORECAST · FP-220** Joint and Marginal Distributions

QUESTION

The joint density is $f(x, y) = cy$ on $0 < y < x < 1$. What is c?

OPTIONS

(a) 2

(b) 3

(c) 6

(d) 4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 6

EXAM SHORTCUT (AI-generated explanation)

The $\int y$ over the triangle is the $\int x^2/2\, dx = 1/6$, so $c = 6$.

54. **FORECAST · FP-221** Joint and Marginal Distributions

QUESTION

For $f(x, y) = 6y$ on $0 < y < x < 1$, the marginal distributions of X and Y are respectively

OPTIONS

(a) $\mathrm{Beta}(3, 1)$ and $\mathrm{Beta}(2, 2)$

(b) $\mathrm{Beta}(2, 1)$ and $\mathrm{Beta}(1, 2)$

(c) $U(0, 1)$ and $\mathrm{Beta}(2, 2)$

(d) $\mathrm{Beta}(3, 1)$ and $\mathrm{Beta}(1, 3)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\mathrm{Beta}(3, 1)$ and $\mathrm{Beta}(2, 2)$

EXAM SHORTCUT (AI-generated explanation)

$f_X = 3x^2$ is $\mathrm{Beta}(3, 1)$ and $f_Y = 6y(1 - y)$ is $\mathrm{Beta}(2, 2)$.

55. **FORECAST · FP-222** Independence and Covariance

QUESTION

For $f(x,y) = 6y$ on $0 < y < x < 1$, what are $\mathrm{P}(X+Y<1)$ and $\mathrm{Corr}(X,Y)$?

OPTIONS

(a) $1/4$ and $1/\sqrt{3}$

(b) $1/2$ and $1/2$

(c) $1/4$ and $1/2$

(d) $1/8$ and $1/\sqrt{3}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/4$ and $1/\sqrt{3}$

EXAM SHORTCUT (AI-generated explanation)

> Two pieces of $1/8$ give $1/4$; $\mathrm{Cov} = 2/5 - 3/8 = 1/40$ with variances $3/80$ and $1/20$ gives $1/\sqrt{3}$.

56. **FORECAST · FP-223** Joint and Marginal Distributions

QUESTION

The joint density is $f(x,y) = ce^{-2x}$ on $0 < y < x < \infty$. What is c?

OPTIONS

(a) 1

(b) 2

(c) 4

(d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 4

EXAM SHORTCUT (AI-generated explanation)

> The $\int xe^{-2x}$ is $1/4$, so $c = 4$.

57. **FORECAST · FP-224** Conditional Distributions

QUESTION

For $f(x, y) = 4e^{-2x}$ on $0 < y < x$, what are the marginal of Y and the conditional law of Y given $X = x$?

OPTIONS

(a) $\mathrm{Exp}(\text{rate } 2)$ and $U(0, x)$

(b) $\Gamma(2, 2)$ and $U(0, x)$

(c) $\mathrm{Exp}(\text{rate } 1)$ and $\mathrm{Exp}(\text{rate } 2)$

(d) $\mathrm{Exp}(\text{rate } 2)$ and $\mathrm{Exp}(\text{rate} x)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\mathrm{Exp}(\text{rate } 2)$ and $U(0, x)$

EXAM SHORTCUT (AI-generated explanation)

> Integrating x from y upwards gives $2e^{-2y}$; dividing by the $\Gamma(2, 2)$ marginal of X leaves the constant $1/x$.

58. **FORECAST · FP-225** Independence and Covariance

QUESTION

For $f(x, y) = 4e^{-2x}$ on $0 < y < x$, what are $\mathrm{Corr}(X, Y)$ and the law of X - Y?

OPTIONS

(a) $1/2$ and $\Gamma(2, 2)$

(b) $1/\sqrt{2}$ and $\mathrm{Exp}(\text{rate } 2)$

(c) $1/\sqrt{2}$ and $U(0, 1)$

(d) $1/\sqrt{3}$ and $\mathrm{Exp}(\text{rate } 2)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/\sqrt{2}$ and $\mathrm{Exp}(\text{rate } 2)$

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{Cov} = 3/4 - 1/2 = 1/4$ with variances $1/2$ and $1/4$, so the correlation is $1/\sqrt{2}$; X - Y is $\mathrm{Exp}(2)$ by symmetry with Y.

59. **FORECAST · FP-226** Joint and Marginal Distributions

QUESTION

The joint pmf is $p(x, y) = c(1/2)^x$ for integers $1 \leq y \leq x$. What is c?

OPTIONS

(a) 1

(b) $1/2$

(c) $1/4$

(d) 2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/2$

EXAM SHORTCUT (AI-generated explanation)

> Summing over y gives $x(1/2)^x$, and the sum of $x(1/2)^x$ over $x \geq 1$ is 2, so $c = 1/2$.

60. **FORECAST · FP-227** Joint and Marginal Distributions

QUESTION

For $p(x, y) = (1/2)(1/2)^x$ on integers $1 \leq y \leq x$, the marginal distribution of Y is

OPTIONS

(a) geometric on {1,2,...} with $p = 1/2$

(b) $\text{Poisson}(1)$

(c) uniform on {1,...,x}

(d) not a standard distribution

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) geometric on {1,2,...} with $p = 1/2$

EXAM SHORTCUT (AI-generated explanation)

> Summing $(1/2)^{x+1}$ over $x \geq y$ gives $(1/2)^y$, the geometric survival pattern.

61. **FORECAST · FP-228** Joint and Marginal Distributions

QUESTION

For $p(x, y) = (1/2)(1/2)^x$ on integers $1 \le y \le x$, what are $\mathrm{P}(X = Y)$ and $\mathrm{E}(X)$?

OPTIONS

(a) $1/2$ and 3

(b) $1/4$ and 2

(c) $1/2$ and 2

(d) $1/3$ and 3

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/2$ and 3

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{P}(X = Y)$ is the sum of $(1/2)^{x+1} = 1/2$, and $\mathrm{E}(X) = (1/2)$ sum of $x^2(1/2)^x = 3$.

62. **FORECAST · FP-229** Compound and Mixture Distributions

QUESTION

The joint density is $f(x, y) = (1/y)e^{-x/y}e^{-y}$ for $x > 0$ and $y > 0$. What are the marginal of Y and the conditional law of X given $Y = y$?

OPTIONS

(a) $\mathrm{Exp}(1)$ and $\mathrm{Exp}(\mathrm{rate}\ 1/y)$

(b) $\Gamma(2, 1)$ and $\mathrm{Exp}(\mathrm{rate} y)$

(c) $\mathrm{Exp}(1)$ and $\mathrm{Exp}(\mathrm{rate} y)$

(d) $U(0, 1)$ and $\mathrm{Exp}(1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\mathrm{Exp}(1)$ and $\mathrm{Exp}(\mathrm{rate}\ 1/y)$

EXAM SHORTCUT (AI-generated explanation)

> Integrating x out leaves e^{-y}, and the first factor is the exponential density with mean y.

63. FORECAST · FP-230 Compound and Mixture Distributions

QUESTION

For $f(x,y) = (1/y)e^{-x/y}e^{-y}$ on the positive quadrant, what is $\mathrm{Var}(X)$?

OPTIONS

(a) 1

(b) 2

(c) 3

(d) 4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 3

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{Var}(X) = \mathrm{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathrm{E}(X \mid Y)) = \mathrm{E}(Y^2) + \mathrm{Var}(Y) = 2 + 1 = 3.$

64. FORECAST · FP-231 Compound and Mixture Distributions

QUESTION

For $f(x,y) = (1/y)e^{-x/y}e^{-y}$ on the positive quadrant, what are $\mathrm{Cov}(X,Y)$ and $\mathrm{P}(X > Y)$?

OPTIONS

(a) 1 and e^{-1}

(b) 0 and $1/2$

(c) 1 and $1/2$

(d) 2 and e^{-1}

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 1 and e^{-1}

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{E}(XY) = \mathrm{E}(Y^2) = 2$ so $\mathrm{Cov} = 1$; and $\mathrm{P}(X > Y \mid Y) = e^{-Y/Y} = e^{-1}$ for every Y.

65. FORECAST · FP-232 Joint and Marginal Distributions

QUESTION

The joint density is $f(x, y) = 1$ on $0 < x < 1$ and $x < y < x + 1$. The marginal of Y is

OPTIONS

(a) $U(0, 2)$

(b) triangular on $(0, 2)$ with peak at 1

(c) $U(1, 2)$

(d) a scaled $\mathrm{Beta}(2, 2)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) triangular on $(0, 2)$ with peak at 1

EXAM SHORTCUT (AI-generated explanation)

> For each y the admissible x range has length y on $(0, 1)$ and 2 - y on $(1, 2)$.

66. FORECAST · FP-233 Independence and Covariance

QUESTION

For $f(x, y) = 1$ on $0 < x < 1, x < y < x + 1$, the variable Y - X is

OPTIONS

(a) $U(0, 1)$ and independent of X

(b) $U(0, 1)$ but dependent on X

(c) triangular

(d) degenerate

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $U(0, 1)$ and independent of X

EXAM SHORTCUT (AI-generated explanation)

> Given $X = x, Y - x$ is $U(0, 1)$ for every x, so the difference is uniform and free of X.

67. **FORECAST · FP-234** Independence and Covariance

QUESTION

For $f(x,y) = 1$ on $0 < x < 1, x < y < x+1$, what is $\mathrm{Corr}(X,Y)$?

OPTIONS

(a) $1/2$

(b) $1/\sqrt{2}$

(c) $1/\sqrt{3}$

(d) 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/\sqrt{2}$

EXAM SHORTCUT (AI-generated explanation)

Writing $Y = X + U$ with U independent $U(0,1)$ gives $\mathrm{Cov} = \mathrm{Var}(X) = 1/12$ and $\mathrm{Var}(Y) = 1/6$.

68. **FORECAST · FP-259** Independence and Covariance

QUESTION

Statement I: if X and Y each have normal marginals then (X, Y) is jointly normal. Statement II: with X standard normal and Y equal to X when $|X|$ is at most c and to -X otherwise, Y is also standard normal. Choose the correct code.

OPTIONS

(a) both true, II explains I

(b) both true, II does not explain I

(c) I true, II false

(d) I false, II true

ANSWER (AI-generated forecast item — no official UPSC key exists)

(d) I false, II true

EXAM SHORTCUT (AI-generated explanation)

Statement II is a construction with normal marginals whose joint law is not normal, refuting Statement I.

69. FORECAST · FP-282 Functions of Random Variables

QUESTION

$X_1, \ldots, X_n$ are independent $\text{Exp}(\lambda)$. What is the probability that X_1 is the smallest?

OPTIONS

(a) $1/n$

(b) $1/2$

(c) $1/\lambda$

(d) $1/n!$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/n$

EXAM SHORTCUT (AI-generated explanation)

> By exchangeability each of the n variables is equally likely to be the minimum.

70. FORECAST · FP-283 Functions of Random Variables

QUESTION

X and Y are independent $\text{Exp}(\lambda)$. Then $|X - Y|$ follows

OPTIONS

(a) $\text{Exp}(\lambda)$

(b) $\text{Exp}(2\lambda)$

(c) $\Gamma(2, \lambda)$

(d) a Laplace distribution

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $\text{Exp}(\lambda)$

EXAM SHORTCUT (AI-generated explanation)

> X - Y is Laplace with scale $1/\lambda$, and folding a Laplace gives back the exponential with the same rate.

71. FORECAST · FP-294 Functions of Random Variables

QUESTION

X and Y are independent $U(0,1)$. What is $\mathrm{P}(X/Y > 2)$?

OPTIONS

(a) $1/4$

(b) $1/3$

(c) $1/2$

(d) $1/8$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/4$

EXAM SHORTCUT (AI-generated explanation)

> The density of X/Y is $1/(2z^2)$ for $z > 1$, and integrating from 2 to ∞ gives $1/4$.

72. FORECAST · FP-295 Functions of Random Variables

QUESTION

X and Y are independent $U(0,1)$ and $Z = XY$. What are the density of Z on $(0,1)$ and $\mathrm{E}(Z)$?

OPTIONS

(a) $-\log z$ and $1/4$

(b) 1 and $1/2$

(c) $2(1-z)$ and $1/3$

(d) $-\log z$ and $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $-\log z$ and $1/4$

EXAM SHORTCUT (AI-generated explanation)

> The product density is $-\log z$, and $\mathrm{E}(Z) = \mathrm{E}(X)\,\mathrm{E}(Y) = 1/4$.

73. **FORECAST · FP-296** Functions of Random Variables

QUESTION

X and Y are independent $\mathrm{Exp}(1)$, with $U = X + Y$ and $V = X - Y$. The joint density of (U, V) on $|v| < u$ is

OPTIONS

(a) e^{-u}

(b) $(1/2)e^{-u}$

(c) e^{-u}/u

(d) $2e^{-u}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $(1/2)e^{-u}$

EXAM SHORTCUT (AI-generated explanation)

> The Jacobian of the inverse map is $1/2$, and the original joint density is $e^{-(x+y)} = e^{-u}$.

74. **FORECAST · FP-297** Independence and Covariance

QUESTION

The covariance matrix of (X_1, X_2) has diagonal entries 2 and off-diagonal entries 1. What is $\mathrm{Corr}(X_1 + X_2, X_1 - X_2)$?

OPTIONS

(a) $1/3$

(b) 0

(c) $1/2$

(d) $1/\sqrt{3}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{Cov}(X_1 + X_2, X_1 - X_2) = \mathrm{Var}(X_1) - \mathrm{Var}(X_2) = 0.$

75. **FORECAST · FP-298** Independence and Covariance

QUESTION

$\mathrm{Corr}(X,Y) = \mathrm{Corr}(X,Z) = 0.9$. What is the feasible range of $\mathrm{Corr}(Y,Z)$?

OPTIONS

(a) -1 to 1

(b) 0.62 to 1

(c) 0.81 to 1

(d) 0.5 to 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0.62 to 1

EXAM SHORTCUT (AI-generated explanation)

> The correlation matrix must be non-negative definite: its determinant condition gives ρ between 0.62 and 1.

76. **FORECAST · FP-299** Independence and Covariance

QUESTION

X and Y are independent each with variance 1. With $Z = X + Y$ and $W = X - 2Y$, what is $\mathrm{Corr}(Z, W)$?

OPTIONS

(a) $-1/\sqrt{10}$

(b) $-1/2$

(c) $1/\sqrt{10}$

(d) $-1/5$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $-1/\sqrt{10}$

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{Cov} = 1 - 2 = -1, \mathrm{Var}(Z) = 2$ and $\mathrm{Var}(W) = 5$, so the correlation is $-1/\sqrt{10}$.

77. FORECAST · FP-300 Conditional Distributions

QUESTION

X and Y are independent geometric on {1,2,...} with the same p. Given $X+Y=n$, the conditional distribution of X is

OPTIONS

(a) $\mathrm{Binomial}(n,1/2)$

(b) uniform on $\{1,\ldots,n-1\}$

(c) geometric

(d) hypergeometric

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) uniform on $\{1,\ldots,n-1\}$

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{P}(X=k,Y=n-k)=p^2q^{n-2}$ is free of k, so all n - 1 admissible values are equally likely.

78. FORECAST · FP-302 Functions of Random Variables

QUESTION

(X, Y) is uniform on the unit disc, with R the distance from the origin and Θ the angle. Which statement is correct?

OPTIONS

(a) R is $U(0,1)$ and Θ is $U(0,2\pi)$, independent

(b) R^2 is $U(0,1)$ and Θ is $U(0,2\pi)$, independent

(c) R is $U(0,1)$ but R and Θ are dependent

(d) R^2 follows $\mathrm{Beta}(2,1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) R^2 is $U(0,1)$ and Θ is $U(0,2\pi)$, independent

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{P}(R\le r)=r^2$, so R^2 is uniform and the angle is independent of it.

79. FORECAST · FP-303 Functions of Random Variables

QUESTION

X is $N(1,1)$ and Y is $N(2,3)$, independent. What is $\mathrm{P}(X < Y)$?

OPTIONS

(a) $\Phi(1)$

(b) $\Phi(1/2)$

(c) $\Phi(1/\sqrt{3})$

(d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\Phi(1/2)$

EXAM SHORTCUT (AI-generated explanation)

Y - X is $N(1,4)$, so $\mathrm{P}(Y - X > 0) = \Phi(1/2)$.

80. FORECAST · FP-304 Compound and Mixture Distributions

QUESTION

The density is $f(x) = (1/2)\phi(x+1) + (1/2)\phi(x-1)$, where ϕ is the standard normal density. What is $\mathrm{Var}(X)$?

OPTIONS

(a) 1

(b) 2

(c) $3/2$

(d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 2

EXAM SHORTCUT (AI-generated explanation)

Within-component variance 1 plus between-component variance 1 gives 2.

81. **FORECAST · FP-305** Functions of Random Variables

QUESTION

X is $U(0,1)$. Then $\min(X, 1-X)$ follows

OPTIONS

(a) $U(0,1/2)$

(b) $\mathrm{Beta}(1,2)$ on $(0,1/2)$

(c) a triangular distribution

(d) $U(0,1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $U(0,1/2)$

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{P(\min > t)} = 1 - 2t$ for $0 < t < 1/2$, which is the $U(0,1/2)$ survival function.

82. **FORECAST · FP-306** Functions of Random Variables

QUESTION

X and Y are independent $U(0,1)$. What is $\mathrm{P}(X^2 + Y^2 \leq 1)$?

OPTIONS

(a) $1/2$

(b) $\pi/4$

(c) $1/\sqrt{2}$

(d) $2/\pi$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\pi/4$

EXAM SHORTCUT (AI-generated explanation)

> The favourable region is the quarter disc of area $\pi/4$ inside the unit square of area 1.

83. **FORECAST · FP-365** Independence and Covariance

QUESTION

Simpson's paradox in a two-way table shows that

OPTIONS

(a) an association present in every subgroup can reverse in the pooled table

(b) correlation always implies causation

(c) the χ^2 statistic is always zero

(d) marginal and conditional distributions always agree

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) an association present in every subgroup can reverse in the pooled table

EXAM SHORTCUT (AI-generated explanation)

> Aggregating over a confounding variable can reverse the direction of an association entirely.

84. **FORECAST · FP-372** Independence and Covariance

QUESTION

Which statement about zero correlation is correct?

OPTIONS

(a) zero correlation always implies independence

(b) independence always implies zero correlation when the variances exist

(c) zero correlation implies independence for any joint distribution with normal marginals

(d) independence and zero correlation are equivalent for all discrete variables

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) independence always implies zero correlation when the variances exist

EXAM SHORTCUT (AI-generated explanation)

> Independence forces $\mathrm{E}(XY) = \mathrm{E}(X)\,\mathrm{E}(Y)$ whenever the moments exist, but the converse needs joint normality.

FORECAST · PROBABILITY

Standard Continuous Distributions

43 forecast problems · 6 concepts

1. FORECAST · FP-021 Exponential Distribution

QUESTION

X follows an exponential distribution with rate θ. Then $\mathrm{Var}(X) > \mathrm{E}(X)$ if and only if

OPTIONS

(a) $\theta > 1$

(b) $\theta < 1$

(c) always

(d) never

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\theta < 1$

EXAM SHORTCUT (AI-generated explanation)

$1/\theta^2 > 1/\theta$ reduces to $\theta < 1$.

2. FORECAST · FP-022 Exponential Distribution

QUESTION

X is $U(0, 1)$ and $Y = -2 \log X$. Then Y follows

OPTIONS

(a) an exponential distribution with mean 1

(b) an exponential distribution with mean 2

(c) $U(0, 2)$

(d) $N(0, 1)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) an exponential distribution with mean 2

EXAM SHORTCUT (AI-generated explanation)

$-\log X$ is $\mathrm{Exp}(1)$, so $-2 \log X$ is Exp with mean 2.

3. **FORECAST · FP-023** Gamma and Beta Distributions

QUESTION

X and Y are independent with X following $\Gamma(3, \lambda)$ and Y following $\Gamma(2, \lambda)$. What is $\mathrm{Var}(X/(X+Y))$?

OPTIONS

(a) $1/25$

(b) $6/25$

(c) $3/5$

(d) $1/30$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/25$

EXAM SHORTCUT (AI-generated explanation)

> $X/(X+Y)$ is $\mathrm{Beta}(3, 2)$, whose variance is $6/(25 \times 6) = 1/25$.

4. **FORECAST · FP-024** Gamma and Beta Distributions

QUESTION

A density is $f(x) = kx^2/(1+x)^6$ for $x > 0$. What is k?

OPTIONS

(a) 20

(b) 30

(c) 60

(d) 120

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 30

EXAM SHORTCUT (AI-generated explanation)

> Beta of the second kind with $a = 3$ and $b = 3$; $B(3, 3) = 1/30$, so $k = 30$.

5. FORECAST · FP-025 Gamma and Beta Distributions

QUESTION

For the density $f(x) = 30x^2/(1+x)^6, x > 0$, what is $\mathrm{E}(X)$?

OPTIONS

(a) 1

(b) $3/2$

(c) 3

(d) does not exist

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $3/2$

EXAM SHORTCUT (AI-generated explanation)

Beta of the second kind with $a = 3, b = 3$: the mean is $a/(b-1) = 3/2$.

6. FORECAST · FP-026 Cauchy and Laplace

QUESTION

X and Y are independent $N(0, 1)$ variables. What is $\mathrm{P}(|X/Y| < 1)$?

OPTIONS

(a) $1/4$

(b) $1/2$

(c) $1/\pi$

(d) $2/\pi$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/2$

EXAM SHORTCUT (AI-generated explanation)

X/Y is standard Cauchy and $\mathrm{P}(|C| < 1) = (2/\pi)\arctan 1 = 1/2$.

7. FORECAST · FP-027 Cauchy and Laplace

QUESTION

X has density $f(x) = (1/2)e^{-|x|}$ on the whole line. What is $\mathrm{P}(|X| > 1)$?

OPTIONS

(a) e^{-1}

(b) $e^{-1}/2$

(c) $1 - e^{-1}$

(d) e^{-2}

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) e^{-1}

EXAM SHORTCUT (AI-generated explanation)

> $|X|$ is $\mathrm{Exp}(1)$ for the standard Laplace, so $\mathrm{P}(|X| > 1) = e^{-1}$.

8. FORECAST · FP-028 Cauchy and Laplace

QUESTION

X has density $f(x) = (1/2)e^{-|x|}$. What is $\mathrm{Var}(X)$?

OPTIONS

(a) 1

(b) 2

(c) $1/2$

(d) 4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 2

EXAM SHORTCUT (AI-generated explanation)

> Laplace with scale $b = 1$ has variance $2b^2 = 2$.

9. FORECAST · FP-029 Lognormal Distribution

QUESTION

If $\log X$ follows $N(0,1)$, what is the median of X?

OPTIONS

(a) $e^{1/2}$

(b) 1

(c) e

(d) 0

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 1

EXAM SHORTCUT (AI-generated explanation)

Medians pass through monotone transformations: the median of $\log X$ is 0, so the median of X is $e^0 = 1$.

10. FORECAST · FP-030 Exponential Distribution

QUESTION

X is exponential with rate θ. What is the maximum over θ of $\mathrm{P}(1 < X < 2)$?

OPTIONS

(a) $1/2$

(b) $1/4$

(c) $1/e$

(d) e^{-2}

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/4$

EXAM SHORTCUT (AI-generated explanation)

With $u = e^{-\theta}$, the probability is $u - u^2$, maximised at $u = 1/2$ giving $1/4$.

11. **FORECAST · FP-031** Normal Distribution

QUESTION

X follows $N(\mu, \sigma^2)$ with $\mathrm{P}(X < 20) = \mathrm{P}(X > 60)$ and $\mathrm{P}(X > 50) = 0.1587$. What is σ?

OPTIONS

(a) 5

(b) 10

(c) 20

(d) 15

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 10

EXAM SHORTCUT (AI-generated explanation)

Symmetry gives $\mu = 40$; then 0.1587 means 50 is one standard deviation above 40, so $\sigma = 10$.

12. **FORECAST · FP-032** Gamma and Beta Distributions

QUESTION

What is the mode of the Beta distribution of the first kind with parameters 2 and 3?

OPTIONS

(a) $2/5$

(b) $1/3$

(c) $1/2$

(d) $2/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/3$

EXAM SHORTCUT (AI-generated explanation)

Mode = $(a - 1)/(a + b - 2) = 1/3$.

13. FORECAST · FP-034 Uniform Distribution

QUESTION

X is $U(a, b)$ with mean 5 and variance 3. What is b?

OPTIONS

(a) 8

(b) 6

(c) $5 + \sqrt{3}$

(d) 11

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 8

EXAM SHORTCUT (AI-generated explanation)

$(b - a)^2 = 36$ gives $b - a = 6$, and $a + b = 10$, so $b = 8$.

14. FORECAST · FP-125 Exponential Distribution

QUESTION

X is exponential with rate 2. What is $\mathrm{E}(X \mid X > 3)$?

OPTIONS

(a) 3

(b) 3.5

(c) 5

(d) 0.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 3.5

EXAM SHORTCUT (AI-generated explanation)

Memorylessness: $\mathrm{E}(X \mid X > a) = a + 1$/rate = $3 + 0.5 = 3.5$.

15. **FORECAST · FP-126** Gamma and Beta Distributions

QUESTION

X follows Γ with shape 3 and rate 2. What is $\mathrm{E}(1/X)$?

OPTIONS

(a) $2/3$

(b) 1

(c) $1/2$

(d) it does not exist

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 1

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{E}(1/X)$ = rate/(shape - 1) = $2/2 = 1$.

16. **FORECAST · FP-127** Lognormal Distribution

QUESTION

If $\log X$ follows $N(0, 1)$, what is $\mathrm{E}(X^2)$?

OPTIONS

(a) e

(b) e^2

(c) $e^{1/2}$

(d) $e^2 - e$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) e^2

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{E}(X^k) = \exp(k\mu + k^2\sigma^2/2)$, so $\mathrm{E}(X^2) = e^2$.

17. **FORECAST · FP-128** Gamma and Beta Distributions

QUESTION

X follows Beta of the first kind with parameters a and b. What is $\mathrm{E}[X(1-X)]$?

OPTIONS

(a) ab/$(a+b)^2$

(b) ab/$((a+b)(a+b+1))$

(c) the variance of X

(d) ab/$((a+b)^2(a+b+1))$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) ab/$((a+b)(a+b+1))$

EXAM SHORTCUT (AI-generated explanation)

> The ratio $B(a+1,b+1)/B(a,b)$ equals ab/$((a+b)(a+b+1))$.

18. **FORECAST · FP-129** Normal Distribution

QUESTION

X follows $N(0,1)$. What is $\mathrm{E}(X^4)$?

OPTIONS

(a) 1

(b) 2

(c) 3

(d) 6

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 3

EXAM SHORTCUT (AI-generated explanation)

> The standard normal moments are $1, 3, 15, \ldots$: $\mathrm{E}(X^{2k}) = (2k-1)!!$ so $\mathrm{E}(X^4) = 3$.

19. FORECAST · FP-130 Cauchy and Laplace

QUESTION

X is standard Cauchy. Which of the following is NOT also standard Cauchy?

OPTIONS

(a) $1/X$

(b) -X

(c) the average of X and an independent copy

(d) X^2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(d) X^2

EXAM SHORTCUT (AI-generated explanation)

The Cauchy is closed under reciprocal, sign change and averaging, but its square is a positive variable and cannot be Cauchy.

20. FORECAST · FP-131 Cauchy and Laplace

QUESTION

X follows $\text{Laplace}(0, b)$. Then $|X|$ follows

OPTIONS

(a) an exponential with rate b

(b) an exponential with rate $1/b$

(c) $\Gamma(2, 1/b)$

(d) a half-normal distribution

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) an exponential with rate $1/b$

EXAM SHORTCUT (AI-generated explanation)

Folding the Laplace gives the exponential with the same scale b, i.e. rate $1/b$.

21. FORECAST · FP-132 Uniform Distribution

QUESTION

X is $U(0,1)$. What is $\mathrm{Var}(X^2)$?

OPTIONS

(a) $1/12$

(b) $4/45$

(c) $1/45$

(d) $1/5$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $4/45$

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{E}(X^4) - [\mathrm{E}(X^2)]^2 = 1/5 - 1/9 = 4/45.$

22. FORECAST · FP-134 Gamma and Beta Distributions

QUESTION

What is the mode of the Γ distribution with shape 2 and rate 1?

OPTIONS

(a) 2

(b) 1

(c) 0

(d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 1

EXAM SHORTCUT (AI-generated explanation)

The Γ mode is (shape - 1)/rate = 1.

23. FORECAST · FP-135 Normal Distribution

QUESTION

For X following $N(\mu, \sigma^2)$, what is $\mathrm{P}(|X - \mu| < 0.6745\sigma)$?

OPTIONS

(a) 0.6827

(b) 0.5

(c) 0.75

(d) 0.9

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0.5

EXAM SHORTCUT (AI-generated explanation)

> 0.6745σ is the quartile deviation, so the interval covers exactly the middle half.

24. FORECAST · FP-136 Gamma and Beta Distributions

QUESTION

X follows Beta of the first kind with parameters 2 and 3, and $Y = X/(1 - X)$. What is $\mathrm{E}(Y)$?

OPTIONS

(a) $2/3$

(b) 1

(c) 2

(d) $1/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 1

EXAM SHORTCUT (AI-generated explanation)

> The odds transform of Beta $I(a, b)$ is Beta II(a,b), whose mean is $a/(b - 1) = 2/2 = 1$.

25. FORECAST · FP-137 Uniform Distribution

QUESTION

X is $U(0, 1)$. Compare $\mathrm{P}(X > 0.7 \mid X > 0.5)$ with $\mathrm{P}(X > 0.2)$.

OPTIONS

(a) 0.6 and 0.8

(b) 0.8 and 0.8

(c) 0.7 and 0.2

(d) 0.3 and 0.8

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 0.6 and 0.8

EXAM SHORTCUT (AI-generated explanation)

The conditional probability is $0.3/0.5 = 0.6$, whereas $\mathrm{P}(X > 0.2) = 0.8$; the uniform is not memoryless.

26. FORECAST · FP-138 Lognormal Distribution

QUESTION

For the lognormal distribution, the correct ordering of mode, median and mean is

OPTIONS

(a) mean < median < mode

(b) mode < median < mean

(c) all three are equal

(d) median < mode < mean

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) mode < median < mean

EXAM SHORTCUT (AI-generated explanation)

They are $e^{\mu-\sigma^2}$, e^{μ} and $e^{\mu+\sigma^2/2}$, which increase in that order.

27. FORECAST · FP-238 Gamma and Beta Distributions

QUESTION

A density is $f(x) = k\sqrt{x}/(1+x)^3$ for $x > 0$. What is k?

OPTIONS

(a) $8/\pi$

(b) $4/\pi$

(c) $2/\pi$

(d) $16/\pi$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $8/\pi$

EXAM SHORTCUT (AI-generated explanation)

Beta II with $a = 3/2$ and $b = 3/2$; $B(3/2, 3/2) = \pi/8$, so $k = 8/\pi$.

28. FORECAST · FP-239 Gamma and Beta Distributions

QUESTION

For the density $f(x)$ proportional to $\sqrt{x}/(1+x)^3$ on $x > 0$, the mean and the variance are

OPTIONS

(a) 3 and 3

(b) 3 and non-existent

(c) 1 and non-existent

(d) $3/2$ and $3/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 3 and non-existent

EXAM SHORTCUT (AI-generated explanation)

Beta II$(3/2, 3/2)$: the mean is $a/(b-1) = 3$, and the variance needs $b > 2$, which fails.

29. FORECAST · FP-240 Gamma and Beta Distributions

QUESTION

A density is $f(x) = k/(\sqrt{x}(1+x))$ for $x > 0$. What are k and the identity of this law?

OPTIONS

(a) $1/\pi$, the square of a standard Cauchy variable

(b) $2/\pi$, the square of a standard normal variable

(c) $1/\pi$, an exponential distribution

(d) $1/2$, Beta of the second kind with parameters 1 and 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/\pi$, the square of a standard Cauchy variable

EXAM SHORTCUT (AI-generated explanation)

Beta II$(1/2, 1/2)$ with $B(1/2, 1/2) = \pi$, and $\tan^2$ of a uniform angle gives exactly this law.

30. FORECAST · FP-241 Gamma and Beta Distributions

QUESTION

For the density $f(x) = 1/(\pi\sqrt{x}(1+x))$ on $x > 0$, what is $\mathrm{P}(X > 1)$?

OPTIONS

(a) $1/4$

(b) $1/2$

(c) $1/\pi$

(d) $1/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/2$

EXAM SHORTCUT (AI-generated explanation)

X is the square of a standard Cauchy C, and $\mathrm{P}(|C| > 1) = 1/2$.

31. FORECAST · FP-242 Gamma and Beta Distributions

QUESTION

A density is $f(x) = k/(\sqrt{x}(1+x)^2)$ for $x > 0$. What are k and $\mathrm{E}(X)$?

OPTIONS

(a) $2/\pi$ and 1

(b) $1/\pi$ and 1

(c) $2/\pi$ and $1/2$

(d) $4/\pi$ and 2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $2/\pi$ and 1

EXAM SHORTCUT (AI-generated explanation)

Beta II$(1/2, 3/2)$: $B = \sqrt{\pi}(\sqrt{\pi}/2) = \pi/2$ so $k = 2/\pi$, and the mean is $(1/2)/(1/2) = 1$.

32. FORECAST · FP-243 Gamma and Beta Distributions

QUESTION

A density is $f(x) = kx^2/(1+3x)^5$ for $x > 0$. What are k and $\mathrm{E}(X)$?

OPTIONS

(a) 27 and 1

(b) 324 and 1

(c) 108 and $1/3$

(d) 324 and 3

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 324 and 1

EXAM SHORTCUT (AI-generated explanation)

Substituting $u = 3x$ turns it into Beta II(3,2); the integral is $B(3,2)/27 = 1/324$, so $k = 324$ and $\mathrm{E}(X) = \mathrm{E}(U)/3 = 1$.

33. FORECAST · FP-284 Normal Distribution

QUESTION

X is $N(0,1)$. What are E[X times the indicator that $X > 0$] and $\mathrm{E}(X \mid X > 0)$?

OPTIONS

(a) $1/\sqrt{2\pi}$ and $\sqrt{2/\pi}$

(b) $1/2$ and 1

(c) $\sqrt{2/\pi}$ and $\sqrt{2/\pi}$

(d) $1/\sqrt{2\pi}$ and $1/\sqrt{2\pi}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/\sqrt{2\pi}$ and $\sqrt{2/\pi}$

EXAM SHORTCUT (AI-generated explanation)

The truncated integral equals the density height at 0, and dividing by $\mathrm{P}(X > 0) = 1/2$ doubles it.

34. FORECAST · FP-285 Normal Distribution

QUESTION

X is $N(\mu, \sigma^2)$. What is $\mathrm{P(X > mu + sigma\ given\ X > mu)}$?

OPTIONS

(a) 0.1587

(b) 0.3174

(c) 0.6826

(d) 0.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 0.3174

EXAM SHORTCUT (AI-generated explanation)

$0.1587/0.5 = 0.3174.$

35. **FORECAST · FP-286** Uniform Distribution

QUESTION

X is $U(0,1)$. What is $\mathrm{Var}(|X-1/2|)$?

OPTIONS

(a) $1/12$

(b) $1/48$

(c) $1/24$

(d) $1/16$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/48$

EXAM SHORTCUT (AI-generated explanation)

$|X-1/2|$ is $U(0,1/2)$, whose variance is $(1/2)^2/12 = 1/48$.

36. **FORECAST · FP-287** Gamma and Beta Distributions

QUESTION

For the Γ distribution with shape α and rate λ, the coefficient of variation and the skewness are

OPTIONS

(a) $1/\sqrt{\alpha}$ and $2/\sqrt{\alpha}$

(b) $1/\alpha$ and $2/\alpha$

(c) $\sqrt{\alpha}$ and $2\sqrt{\alpha}$

(d) $1/\sqrt{\alpha}$ and $1/\sqrt{\alpha}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $1/\sqrt{\alpha}$ and $2/\sqrt{\alpha}$

EXAM SHORTCUT (AI-generated explanation)

Both shape measures depend only on α: $\mathrm{CV} = 1/\sqrt{\alpha}$ and skewness = $2/\sqrt{\alpha}$.

37. **FORECAST · FP-288** Gamma and Beta Distributions

QUESTION

What is the variance of the Beta distribution with both parameters equal to $1/2$ (the arcsine law)?

OPTIONS

(a) $1/12$

(b) $1/8$

(c) $1/4$

(d) $1/16$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/8$

EXAM SHORTCUT (AI-generated explanation)

ab/$((a+b)^2(a+b+1)) = (1/4)/(1 \times 2) = 1/8.$

38. **FORECAST · FP-289** Cauchy and Laplace

QUESTION

X follows the Cauchy distribution centred at θ with scale 1. What is $\mathrm{P}(|X-\theta| > \sqrt{3})$?

OPTIONS

(a) $1/4$

(b) $1/3$

(c) $1/2$

(d) $1/\sqrt{3}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $1/3$

EXAM SHORTCUT (AI-generated explanation)

$1-(2/\pi)\arctan(\sqrt{3}) = 1-(2/\pi)(\pi/3) = 1/3.$

39. FORECAST · FP-290 Cauchy and Laplace

QUESTION

For the Laplace distribution with scale b, what is the kurtosis measured as the fourth central moment over the squared variance?

OPTIONS

(a) 3

(b) 6

(c) 9

(d) 4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 6

EXAM SHORTCUT (AI-generated explanation)

$\mu_4 = 24b^4$ and $\sigma^2 = 2b^2$, so the ratio is $24/4 = 6$.

40. FORECAST · FP-291 Exponential Distribution

QUESTION

X is $\mathrm{Exp}(\lambda)$. The fractional part of X has density on $(0, 1)$ equal to

OPTIONS

(a) $\lambda e^{-\lambda y}$

(b) $\lambda e^{-\lambda y}/(1 - e^{-\lambda})$

(c) 1

(d) $e^{-\lambda y}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\lambda e^{-\lambda y}/(1 - e^{-\lambda})$

EXAM SHORTCUT (AI-generated explanation)

Summing the shifted pieces gives $\lambda e^{-\lambda y}$ times the geometric sum $1/(1 - e^{-\lambda})$.

41. **FORECAST · FP-292** Normal Distribution

QUESTION

X and Y are independent $N(0,1)$. What is $\mathrm{E}|X-Y|$?

OPTIONS

(a) $\sqrt{2/\pi}$

(b) $2/\sqrt{\pi}$

(c) 1

(d) $\sqrt{2}$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $2/\sqrt{\pi}$

EXAM SHORTCUT (AI-generated explanation)

X - Y is $N(0,2)$, and $\mathrm{E}|Z| = \sigma\sqrt{2/\pi} = \sqrt{2}\sqrt{2/\pi} = 2/\sqrt{\pi}$.

42. **FORECAST · FP-293** Lognormal Distribution

QUESTION

Consider: products of independent lognormals are lognormal, and sums of independent lognormals are lognormal. These statements are

OPTIONS

(a) both true

(b) the first true, the second false

(c) the first false, the second true

(d) both false

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the first true, the second false

EXAM SHORTCUT (AI-generated explanation)

Logs of products add, so products stay lognormal; sums have no such structure and are not lognormal.

43. FORECAST · FP-371 Normal Distribution

QUESTION

The memoryless property characterises

OPTIONS

(a) the normal distribution among continuous laws

(b) the exponential among continuous laws and the geometric among discrete laws

(c) the uniform distribution

(d) every distribution with a constant hazard rate except the exponential

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the exponential among continuous laws and the geometric among discrete laws

EXAM SHORTCUT (AI-generated explanation)

A constant hazard rate is equivalent to memorylessness, and only the exponential and the geometric have it.

FORECAST · PROBABILITY

Standard Discrete Distributions

30 forecast problems · 5 concepts

1. FORECAST · FP-011 Binomial Distribution

QUESTION

What is the mode of the $\mathrm{Binomial}(15, 0.4)$ distribution?

OPTIONS

(a) 5

(b) 6

(c) 7

(d) 6 and 7

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 6

EXAM SHORTCUT (AI-generated explanation)

$(n+1)p = 16(0.4) = 6.4$, not an integer, so the mode is $\mathrm{floor}(6.4) = 6$ and it is unique.

2. FORECAST · FP-012 Poisson Distribution

QUESTION

X follows a Poisson distribution with $\mathrm{P}(X=2) = \mathrm{P}(X=3)$. What is $\mathrm{P}(X=0)$?

OPTIONS

(a) e^{-2}

(b) e^{-3}

(c) $3e^{-3}$

(d) e^{-1}

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) e^{-3}

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{P}(X=3)/\mathrm{P}(X=2) = \lambda/3 = 1$ gives $\lambda = 3$, so $\mathrm{P}(X=0) = e^{-3}$.

3. FORECAST · FP-013 Geometric and Negative Binomial

QUESTION

X is geometric on $\{1, 2, \ldots\}$ with success probability p and $q = 1 - p$. What is $\mathrm{P}(X > m + n \mid X > m)$?

OPTIONS

(a) q^n

(b) q^m

(c) q^{m+n}

(d) pq^n

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) q^n

EXAM SHORTCUT (AI-generated explanation)

Memorylessness: the conditional law of X - m given $X > m$ is the original geometric law, so the answer is q^n.

4. FORECAST · FP-014 Geometric and Negative Binomial

QUESTION

A fair coin is tossed repeatedly. What is the probability that the 5th head occurs on the 8th toss?

OPTIONS

(a) $35/256$

(b) $21/256$

(c) $35/128$

(d) $56/256$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) $35/256$

EXAM SHORTCUT (AI-generated explanation)

Negative binomial: $\binom{7}{4}(1/2)^8 = 35/256$.

5. **FORECAST · FP-015** Multinomial Distribution

QUESTION

In a multinomial experiment with $n = 10$ and cell probabilities $(0.2, 0.3, 0.5)$, what is $\mathrm{Cov}(X_1, X_2)$?

OPTIONS

(a) 0.6

(b) -0.6

(c) -0.06

(d) 0

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) -0.6

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{Cov} = -np_1p_2 = -10(0.2)(0.3) = -0.6.$

6. **FORECAST · FP-016** Hypergeometric Distribution

QUESTION

For a hypergeometric distribution with $N = 20, K = 8$ and $n = 5$, what is $\mathrm{Var}(X)$?

OPTIONS

(a) $6/5$

(b) $18/19$

(c) $24/19$

(d) 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $18/19$

EXAM SHORTCUT (AI-generated explanation)

> Binomial variance $5(0.4)(0.6) = 1.2$ times the finite-population correction $15/19$ gives $18/19$.

7. FORECAST · FP-017 Binomial Distribution

QUESTION

A random variable has pmf $p(x) = \binom{8}{x} 3^x 5^{8-x}/8^8$ for $x = 0, 1, \ldots, 8$. What is $\mathrm{Var}(X)$?

OPTIONS

(a) 3

(b) $15/8$

(c) $5/8$

(d) $24/8$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $15/8$

EXAM SHORTCUT (AI-generated explanation)

Write $8^8 = (3+5)^8$ to recognise $\mathrm{Binomial}(8, 3/8)$; then $npq = 8(3/8)(5/8) = 15/8$.

8. FORECAST · FP-018 Binomial Distribution

QUESTION

X is $\mathrm{Binomial}(n, p)$ with $\mathrm{E}(X) = 6$ and $\mathrm{Var}(X) = 4.2$. What is n?

OPTIONS

(a) 10

(b) 15

(c) 20

(d) 30

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) 20

EXAM SHORTCUT (AI-generated explanation)

$q = \mathrm{Var}$/mean = $4.2/6 = 0.7$, so $p = 0.3$ and $n = 6/0.3 = 20$.

9. FORECAST · FP-019 Poisson Distribution

QUESTION

X and Y are independent Poisson variables with means λ_1 and λ_2. The conditional distribution of X given $X + Y = n$ is

OPTIONS

(a) Poisson

(b) $\text{Binomial}(n, \lambda_1/(\lambda_1 + \lambda_2))$

(c) Geometric

(d) Hypergeometric

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\text{Binomial}(n, \lambda_1/(\lambda_1 + \lambda_2))$

EXAM SHORTCUT (AI-generated explanation)

> The Poisson sum is Poisson, and the ratio of the two pmfs collapses to a binomial with $p = \lambda_1/(\lambda_1 + \lambda_2)$.

10. FORECAST · FP-020 Geometric and Negative Binomial

QUESTION

X and Y are independent geometric variables on $\{1, 2, ...\}$ with success probabilities p_1 and p_2. Then $\min(X, Y)$ is geometric with success probability

OPTIONS

(a) $p_1 p_2$

(b) $p_1 + p_2$

(c) $1 - q_1 q_2$

(d) $q_1 q_2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) $1 - q_1 q_2$

EXAM SHORTCUT (AI-generated explanation)

> $\mathrm{P(min > k)} = q_1^k q_2^k = (q_1 q_2)^k$, so the failure probability is $q_1 q_2$ and success is $1 - q_1 q_2$.

11. **FORECAST · FP-113** Binomial Distribution

QUESTION

X is $\text{Binomial}(4, 1/3)$. What is $\text{P}(X \text{ is even})$?

OPTIONS

(a) $1/2$

(b) $41/81$

(c) $40/81$

(d) $17/27$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $41/81$

EXAM SHORTCUT (AI-generated explanation)

> The parity formula gives $[1 + (1 - 2p)^n]/2 = (1 + 1/81)/2 = 41/81$.

12. **FORECAST · FP-114** Poisson Distribution

QUESTION

X is $\text{Poisson}(\lambda)$. What is $\text{E}[X(X - 1)(X - 2)]$?

OPTIONS

(a) $\lambda^3 + 3\lambda^2 + \lambda$

(b) λ^3

(c) $\lambda^3 - \lambda$

(d) λ^2

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) λ^3

EXAM SHORTCUT (AI-generated explanation)

> Factorial moments of the Poisson are pure powers: the r-th factorial moment is λ^r.

13. **FORECAST · FP-115** Poisson Distribution

QUESTION

X is $\mathrm{Poisson}(\lambda)$. What is $\mathrm{P}(X \text{ is even})$?

OPTIONS

(a) $1/2$

(b) $(1+e^{-\lambda})/2$

(c) $(1+e^{-2\lambda})/2$

(d) $e^{-\lambda}\sinh(\lambda)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) $(1+e^{-2\lambda})/2$

EXAM SHORTCUT (AI-generated explanation)

> $[\mathrm{P}(1)+\mathrm{P}(-1)]/2$ with $\mathrm{P}(s)=\exp(\lambda(s-1))$ gives $(1+e^{-2\lambda})/2$.

14. **FORECAST · FP-116** Geometric and Negative Binomial

QUESTION

X is geometric on $\{0,1,2,\ldots\}$ with $p=1/2$. What is $\mathrm{E}(X^2)$?

OPTIONS

(a) 2

(b) 3

(c) 4

(d) 1

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 3

EXAM SHORTCUT (AI-generated explanation)

> Mean 1 and variance $q/p^2=2$, so $\mathrm{E}(X^2)=2+1=3$.

15. FORECAST · FP-117 Geometric and Negative Binomial

QUESTION

For the negative binomial distribution counting failures before the r-th success, the ratio of variance to mean is

OPTIONS

(a) 1

(b) p

(c) $1/p$

(d) r/p

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) $1/p$

EXAM SHORTCUT (AI-generated explanation)

Var/mean = $(rq/p^2)/(rq/p) = 1/p$, which always exceeds 1.

16. FORECAST · FP-118 Hypergeometric Distribution

QUESTION

Comparing $\mathrm{Hypergeometric}(N, K, n)$ with $\mathrm{Binomial}(n, K/N)$, which statement is TRUE?

OPTIONS

(a) the variances are equal

(b) the hypergeometric variance is smaller whenever $n \geq 2$

(c) the hypergeometric variance is larger

(d) it depends on K

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) the hypergeometric variance is smaller whenever $n \geq 2$

EXAM SHORTCUT (AI-generated explanation)

The two differ by the factor $(N - n)/(N - 1)$, which is below 1 as soon as $n \geq 2$.

17. FORECAST · FP-119 Multinomial Distribution

QUESTION

Twelve fair dice are thrown and X_i counts the dice showing face i. What is $\mathrm{Corr}(X_1, X_2)$?

OPTIONS

(a) $-1/6$

(b) $-1/36$

(c) $-1/5$

(d) 0

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) $-1/5$

EXAM SHORTCUT (AI-generated explanation)

The multinomial correlation is $-\sqrt{p_1 p_2/(q_1 q_2)} = -(1/6)/(5/6) = -1/5.$

18. FORECAST · FP-120 Binomial Distribution

QUESTION

X is $\mathrm{Binomial}(2, 1/2)$ and Y is $\mathrm{Binomial}(1, 1/4)$, independent. Is X + Y binomial?

OPTIONS

(a) yes, $\mathrm{Binomial}(3, 5/12)$

(b) yes, $\mathrm{Binomial}(3, 3/8)$

(c) no

(d) yes, $\mathrm{Binomial}(3, 1/2)$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) no

EXAM SHORTCUT (AI-generated explanation)

$\mathrm{P}(S = 0) = 3/16$ and $\mathrm{P}(S = 3) = 1/16$, and no single p can produce both $q^3 = 3/16$ and $p^3 = 1/16$.

19. FORECAST · FP-121 Binomial Distribution

QUESTION

X is $\mathrm{Binomial}(10, p)$ with $\mathrm{P}(X = 3) = \mathrm{P}(X = 4)$. What is p?

OPTIONS

(a) $1/3$

(b) $4/11$

(c) $3/10$

(d) $7/11$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $4/11$

EXAM SHORTCUT (AI-generated explanation)

The ratio condition $(n-k)p = (k+1)q$ with $k = 3$ gives $7p = 4q$, so $p = 4/11$.

20. FORECAST · FP-122 Poisson Distribution

QUESTION

Events occur as a Poisson process of rate λ. Given exactly one event in (0, t], the time of that event is

OPTIONS

(a) exponential truncated to (0, t]

(b) $U(0, t)$

(c) a scaled $\mathrm{Beta}(1, 2)$

(d) degenerate at $t/2$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $U(0, t)$

EXAM SHORTCUT (AI-generated explanation)

Conditioning a Poisson process on one event in an interval makes its position uniform on that interval.

21. **FORECAST · FP-124** Binomial Distribution

QUESTION

X is uniform on $\{1, 2, ..., 7\}$. What is $\mathrm{Var}(X)$?

OPTIONS

(a) 4

(b) $49/12$

(c) $16/3$

(d) 3.5

ANSWER (AI-generated forecast item — no official UPSC key exists)

(a) 4

EXAM SHORTCUT (AI-generated explanation)

> The discrete uniform variance is $(N^2 - 1)/12 = 48/12 = 4$.

22. **FORECAST · FP-274** Binomial Distribution

QUESTION

X is $\mathrm{Binomial}(20, p)$. What is the maximum possible value of $\mathrm{Var}(X)$ over p?

OPTIONS

(a) 20

(b) 5

(c) 10

(d) 4

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) 5

EXAM SHORTCUT (AI-generated explanation)

> npq is maximised at $p = 1/2$, giving $20/4 = 5$.

23. **FORECAST · FP-275** Poisson Distribution

QUESTION

X is $\mathrm{Poisson}(\lambda)$. What is $\mathrm{E}[1/(X+1)]$?

OPTIONS

(a) $1/(\lambda+1)$

(b) $(1-e^{-\lambda})/\lambda$

(c) $e^{-\lambda}$

(d) $1/\lambda$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $(1-e^{-\lambda})/\lambda$

EXAM SHORTCUT (AI-generated explanation)

> Shifting the index turns the sum into $(1/\lambda)$ times the Poisson sum from 1 upwards, giving $(1-e^{-\lambda})/\lambda$.

24. **FORECAST · FP-276** Geometric and Negative Binomial

QUESTION

X and Y are independent geometric variables on {1,2,...} with $p = 1/2$. What is $\mathrm{P}(X+Y=4)$?

OPTIONS

(a) $1/8$

(b) $3/16$

(c) $1/4$

(d) $3/8$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $3/16$

EXAM SHORTCUT (AI-generated explanation)

> The sum is negative binomial: $\binom{3}{1}(1/2)^4 = 3/16$.

25. FORECAST · FP-277 Hypergeometric Distribution

QUESTION

What is the variance of the number of aces in a 13-card bridge hand?

OPTIONS

(a) 1

(b) $12/13$

(c) $12/17$

(d) $3/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(c) $12/17$

EXAM SHORTCUT (AI-generated explanation)

Hypergeometric with $n = 13, K = 4, N = 52$: $13(1/13)(12/13)(39/51) = 12/17$.

26. FORECAST · FP-278 Multinomial Distribution

QUESTION

Twelve fair dice are thrown; X_1 counts the ones and X_2 the twos. Given $X_1 + X_2 = 4$, the conditional distribution of X_1 is

OPTIONS

(a) $\text{Binomial}(4, 1/6)$

(b) $\text{Binomial}(4, 1/2)$

(c) $\text{Binomial}(12, 1/6)$

(d) hypergeometric

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\text{Binomial}(4, 1/2)$

EXAM SHORTCUT (AI-generated explanation)

Conditioning a multinomial on the total of two cells gives a binomial with $p_1/(p_1 + p_2) = 1/2$.

27. FORECAST · FP-279 Geometric and Negative Binomial

QUESTION

Bernoulli trials have success probability $1/3$. What is the probability that the first success occurs on an odd-numbered trial?

OPTIONS

(a) $1/2$

(b) $3/5$

(c) $2/5$

(d) $1/3$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $3/5$

EXAM SHORTCUT (AI-generated explanation)

The $\sum p(1+q^2+q^4+\ldots) = p/(1-q^2) = 1/(1+q) = 3/5$.

28. FORECAST · FP-280 Binomial Distribution

QUESTION

X is $\mathrm{Binomial}(n,p)$ and $Y = n - X$. What is $\mathrm{Corr}(X,Y)$?

OPTIONS

(a) 0

(b) -1

(c) -p

(d) -pq

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) -1

EXAM SHORTCUT (AI-generated explanation)

Y is an exact decreasing linear function of X, so the correlation is -1.

29. FORECAST · FP-281 Poisson Distribution

QUESTION

X is $\text{Poisson}(\lambda)$ and, given X, the variable X_1 is $\text{Binomial}(X, 1/2)$. What is $\text{Var}(X_1)$?

OPTIONS

(a) $\lambda/4$

(b) $\lambda/2$

(c) λ

(d) $\lambda/2 + \lambda^2/4$

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) $\lambda/2$

EXAM SHORTCUT (AI-generated explanation)

> Thinning gives X_1 as $\text{Poisson}(\lambda/2)$, whose variance is $\lambda/2$.

30. FORECAST · FP-366 Poisson Distribution

QUESTION

The Poisson approximation to the $\text{Binomial}(n, p)$ requires

OPTIONS

(a) n large and p fixed

(b) n large, p small, with np moderate

(c) n small and p large

(d) only that np is an integer

ANSWER (AI-generated forecast item — no official UPSC key exists)

(b) n large, p small, with np moderate

EXAM SHORTCUT (AI-generated explanation)

> The rare-event regime keeps np fixed while n grows and p shrinks.